

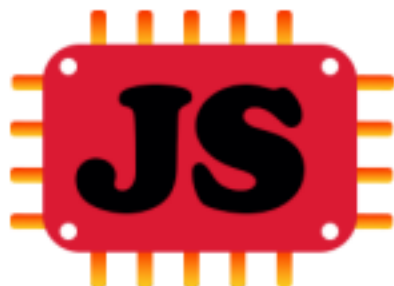
Electronics (BJT)

Lecture 08, 09

Md. Najmus Sakib,

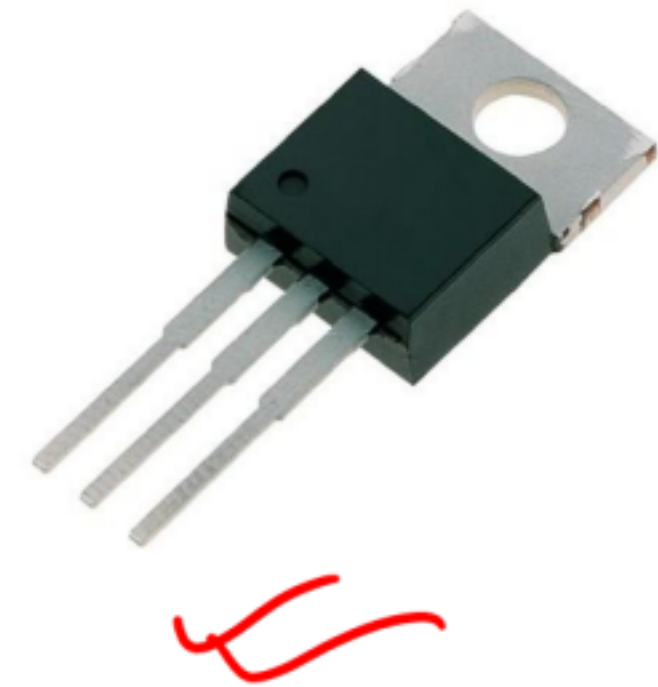
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Assistant Engineer, DWASA.

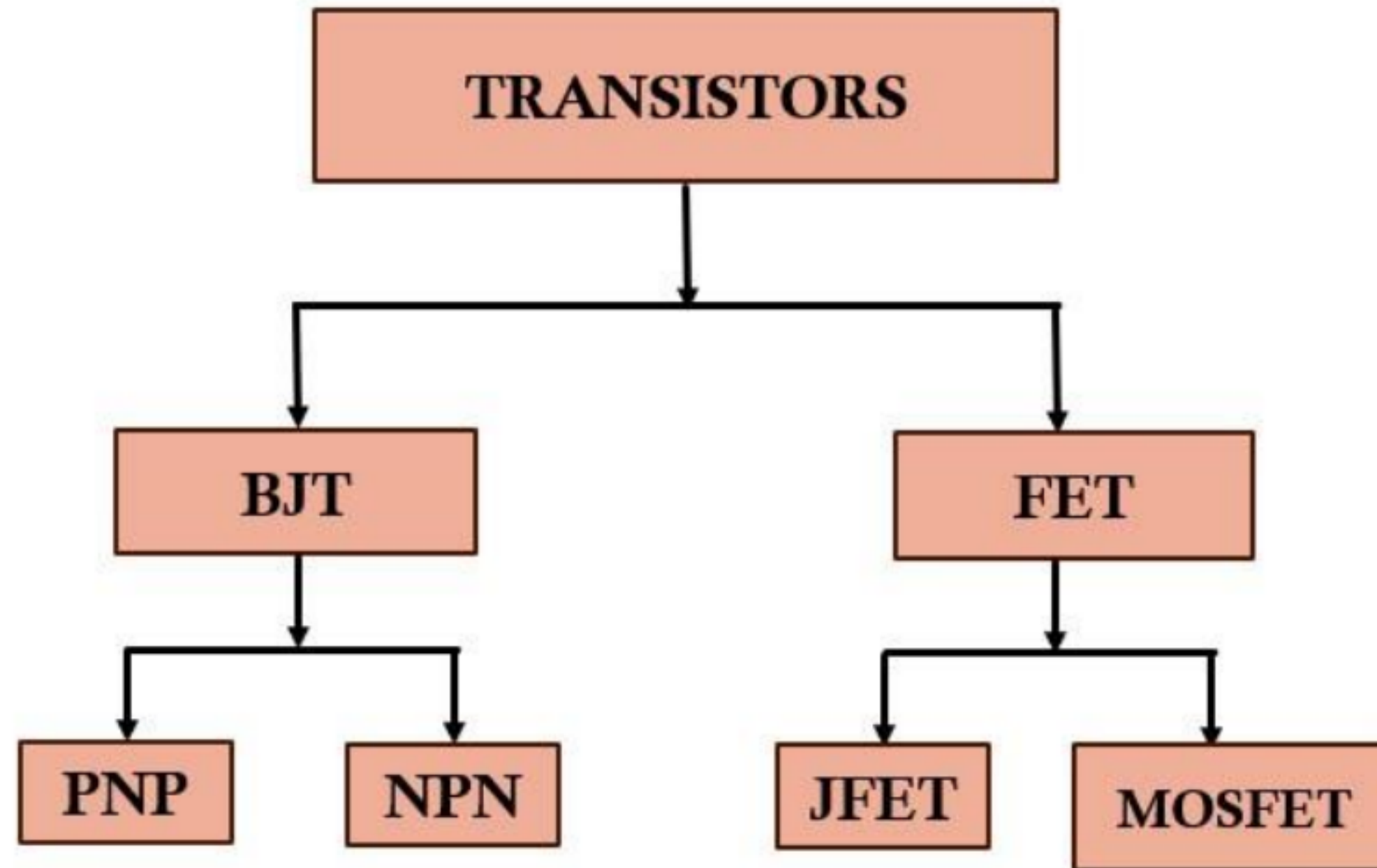


BJT

(Bipolar Junction Transistor)



Types of Transistors



Transistor types

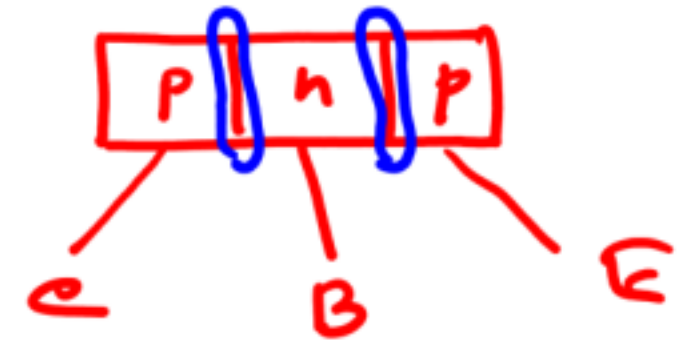
	BJT	FET
1	Bipolar Junction Transistor	Field Effect Transistor
2	Two types: PNP & NPN	Two types: JFET & MOSFET
3	BJT is a current-controlled device	FET is a voltage-controlled device
4	current flows due to both majority and minority charge carriers. That's why, it's a <u>Bi</u> -polar device	current flows due to majority charge carriers only. That's why, it's a <u>Uni</u> -polar device.
5	BJT has a bigger size.	FET has a smaller size
	BJT has three configurations namely common-base, common-emitter, and common-collector	FET has three configurations namely, common-gate, common-source, and common-drain.

BJT Construction

- The BJT is a three-layer semiconductor device.

- 3 layers: emitter, base, collector

- 2 junctions: c-B, B-E

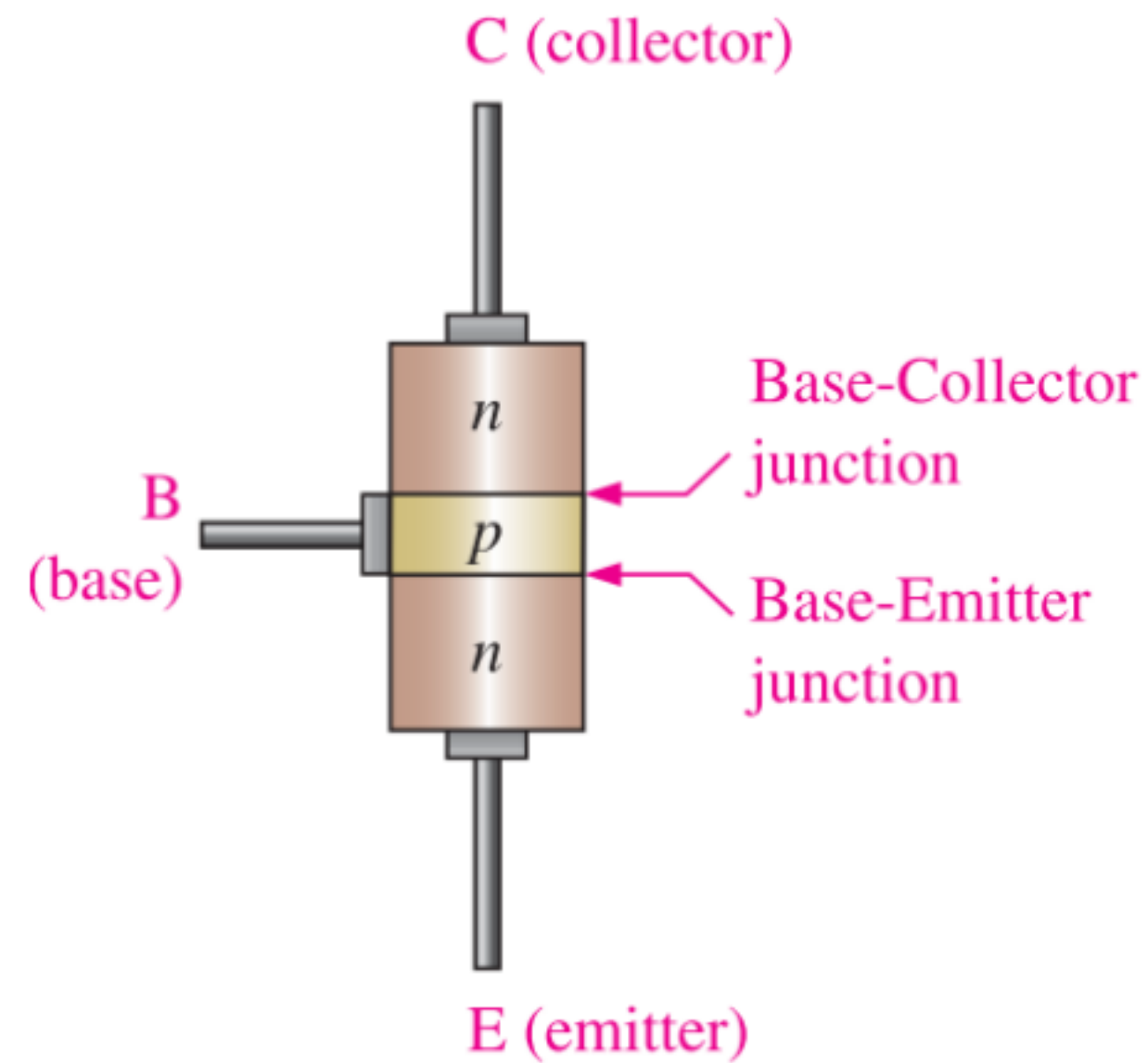
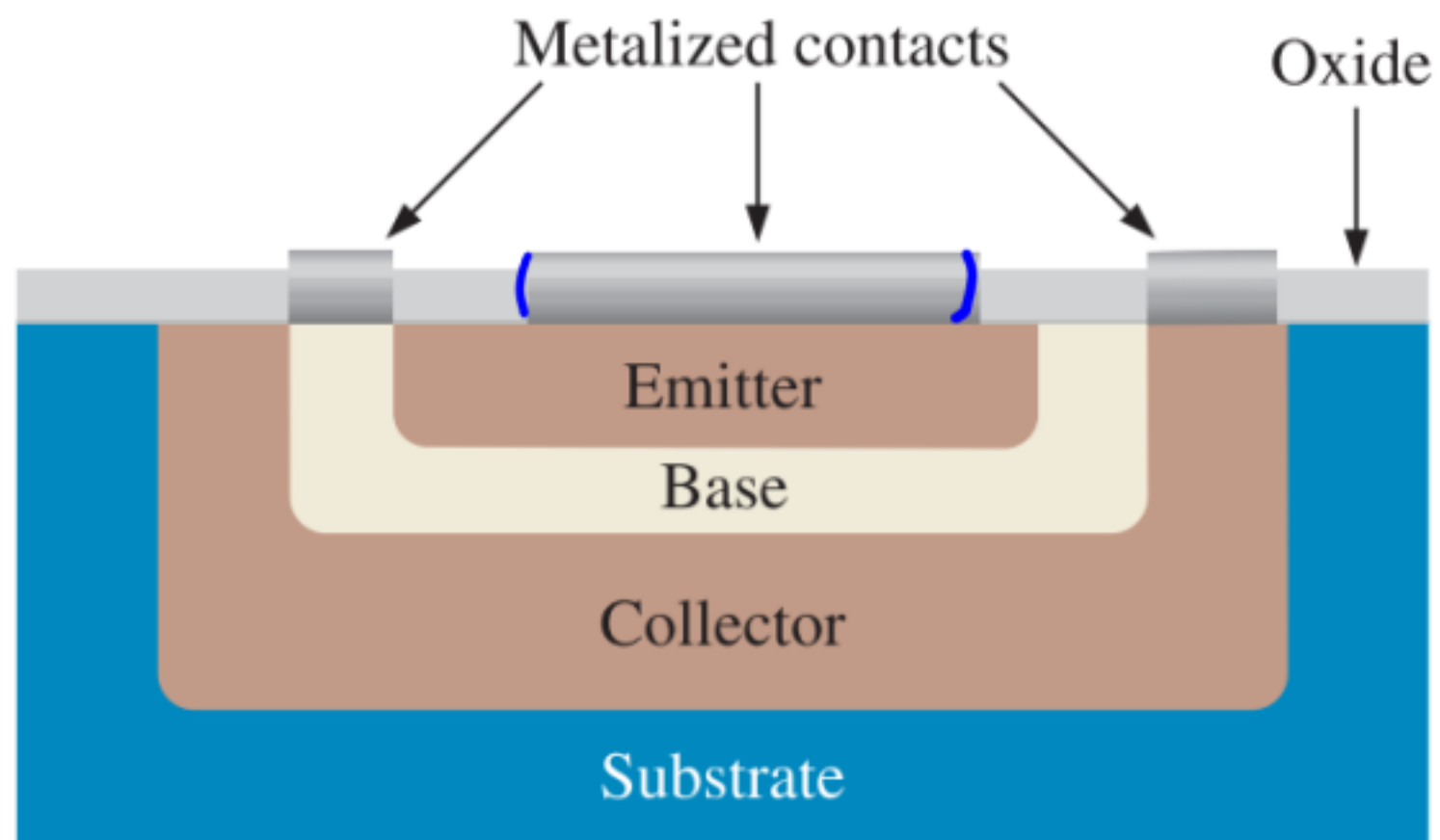


- Layers are doped with either p-type or n-type semiconductors

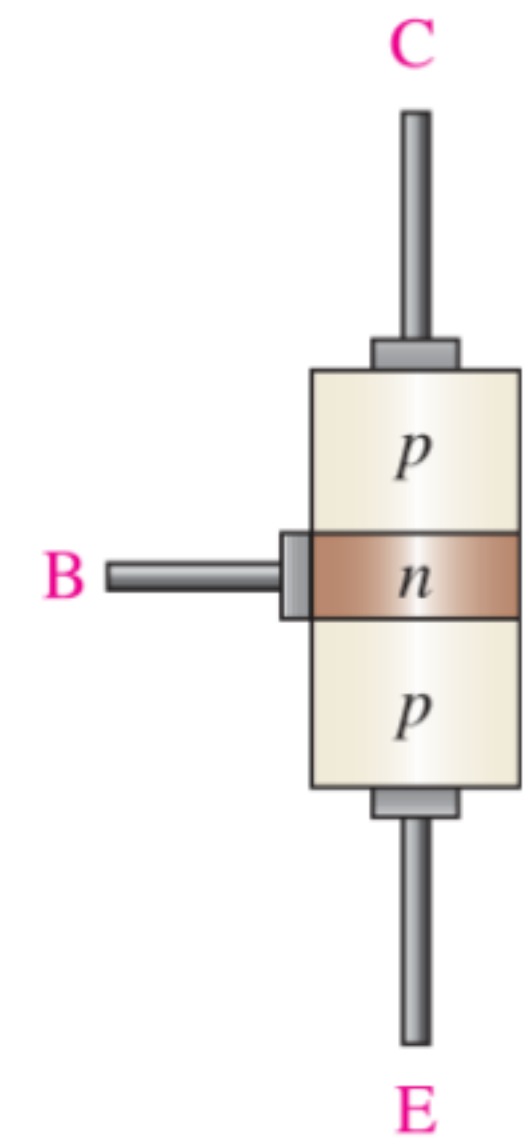
Type 01: two n- and one p-type layers of material > npn transistor

Type 02: two p- and one n-type layers of material > pnp transistor

BJT structure

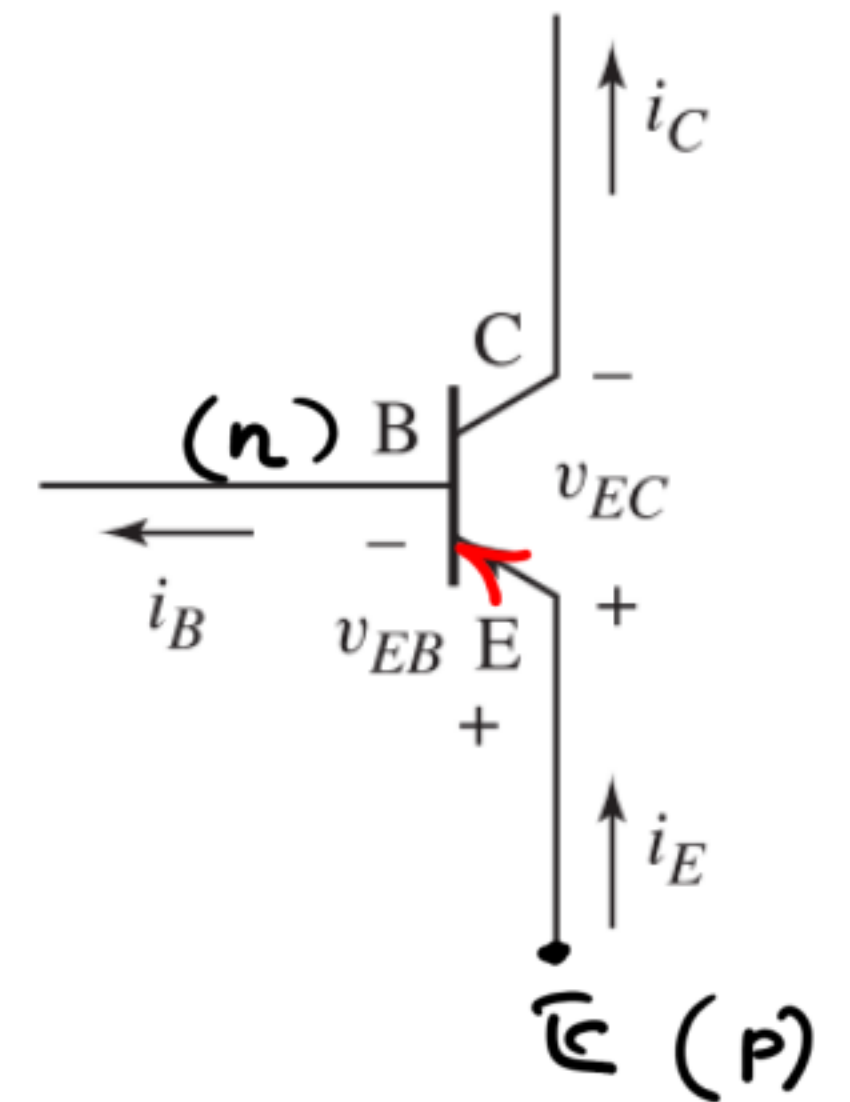
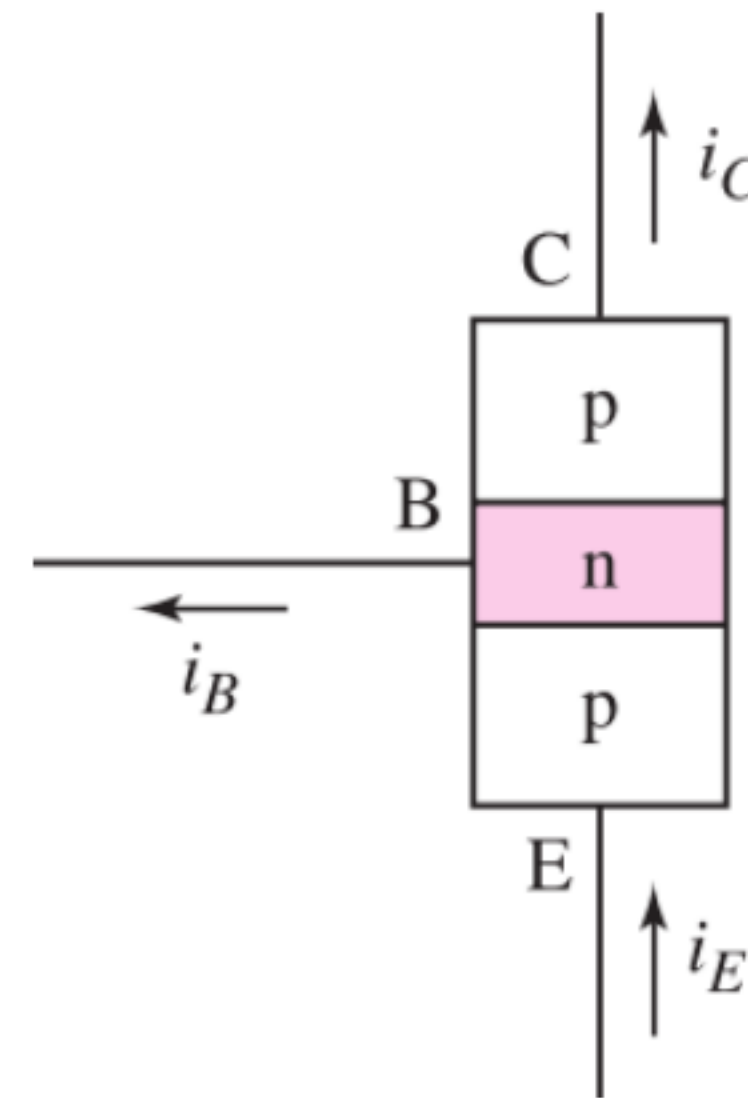
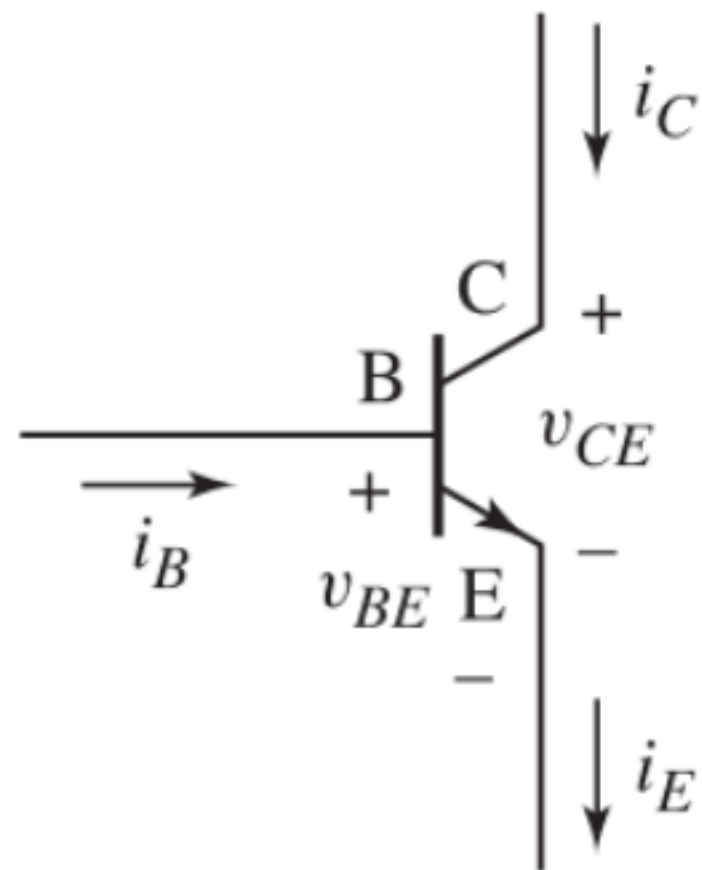
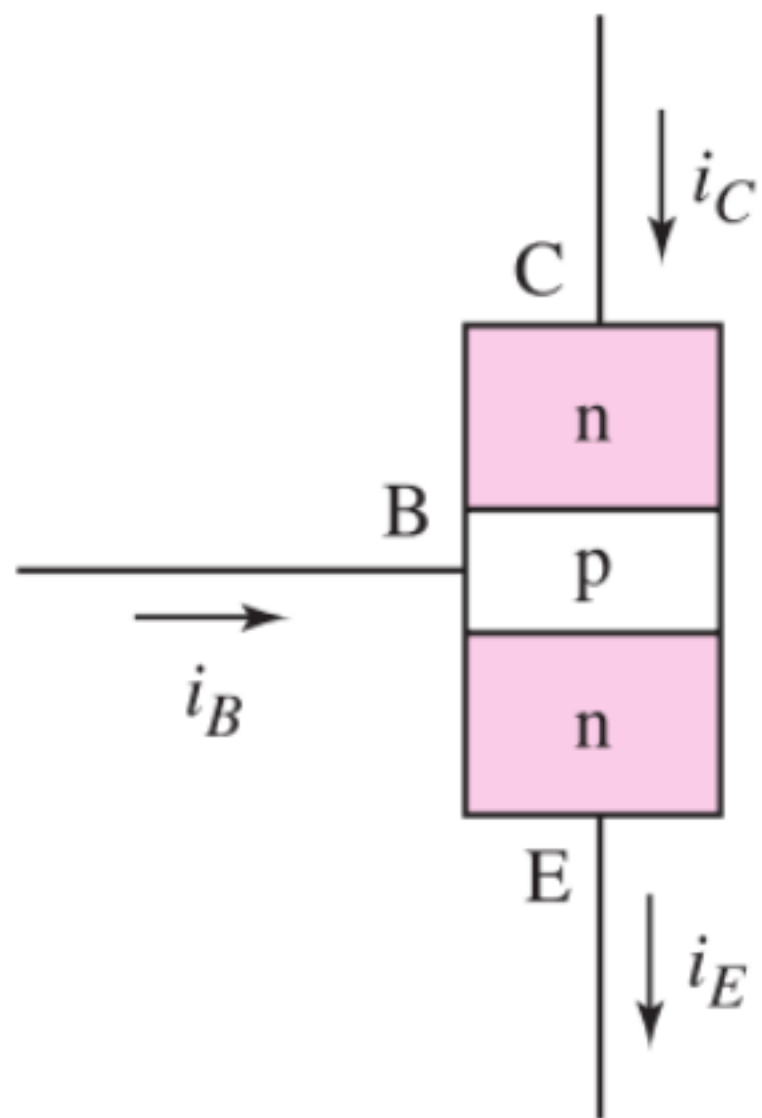


(b) *nnp*



(c) *pnp*

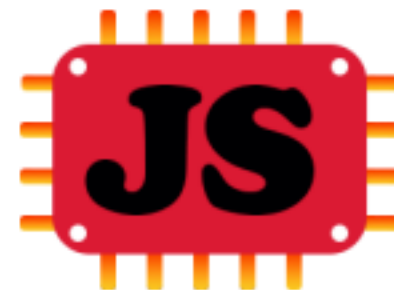
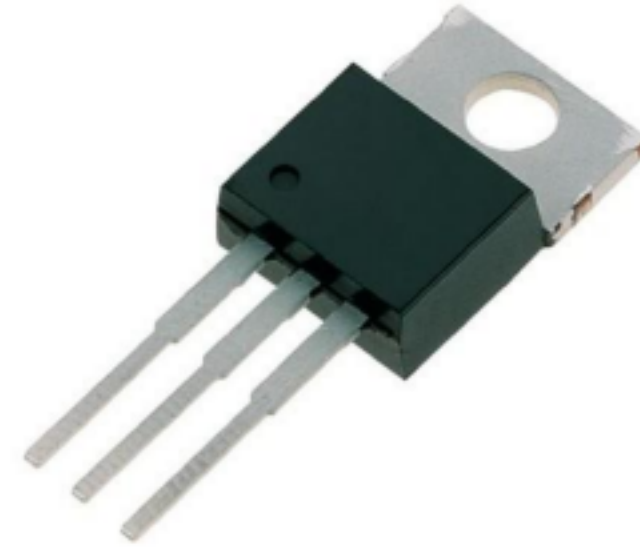
Symbology & Current flow in BJT



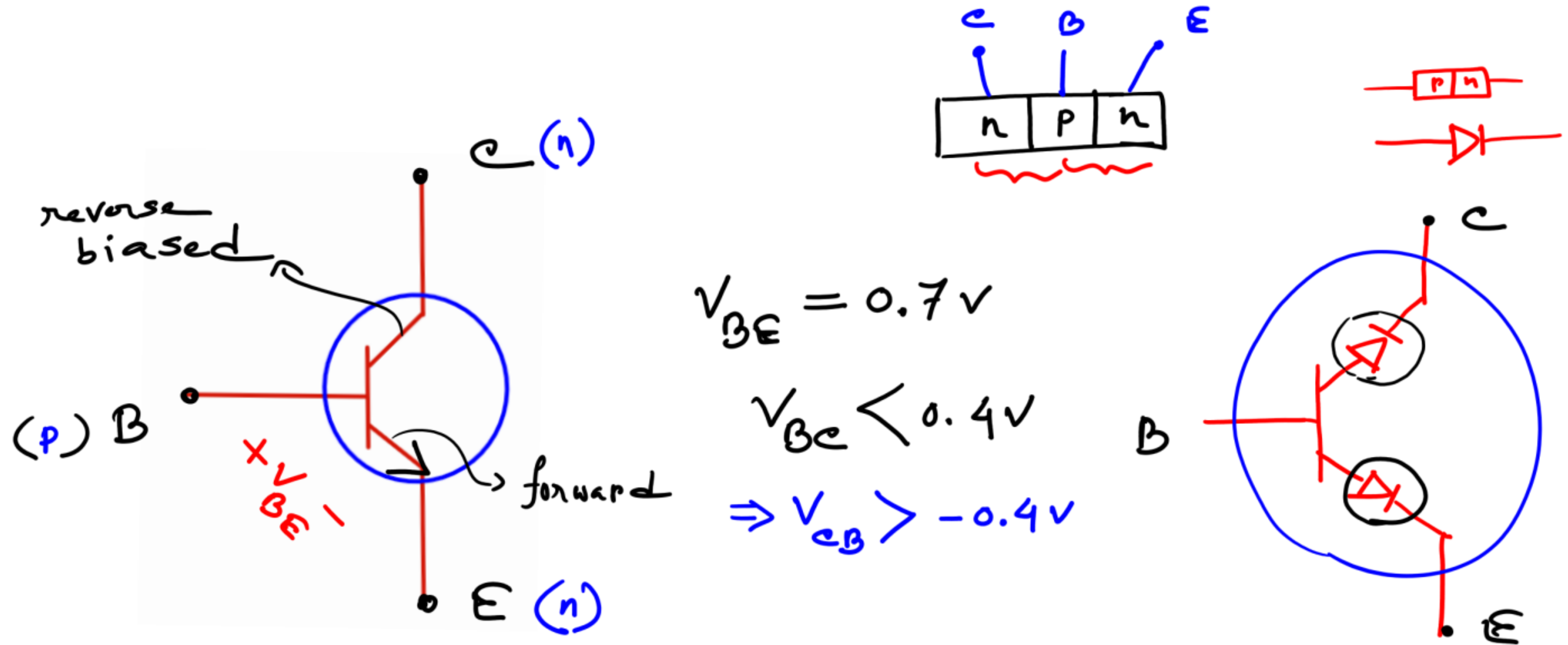
Modes of Operation

- 4 modes of operation

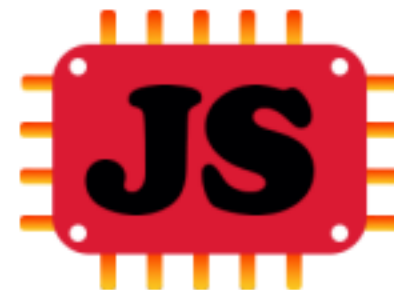
1. Active → *amplifier*
2. Saturation } → *switch*
3. Cut-off }
4. Reverse-active



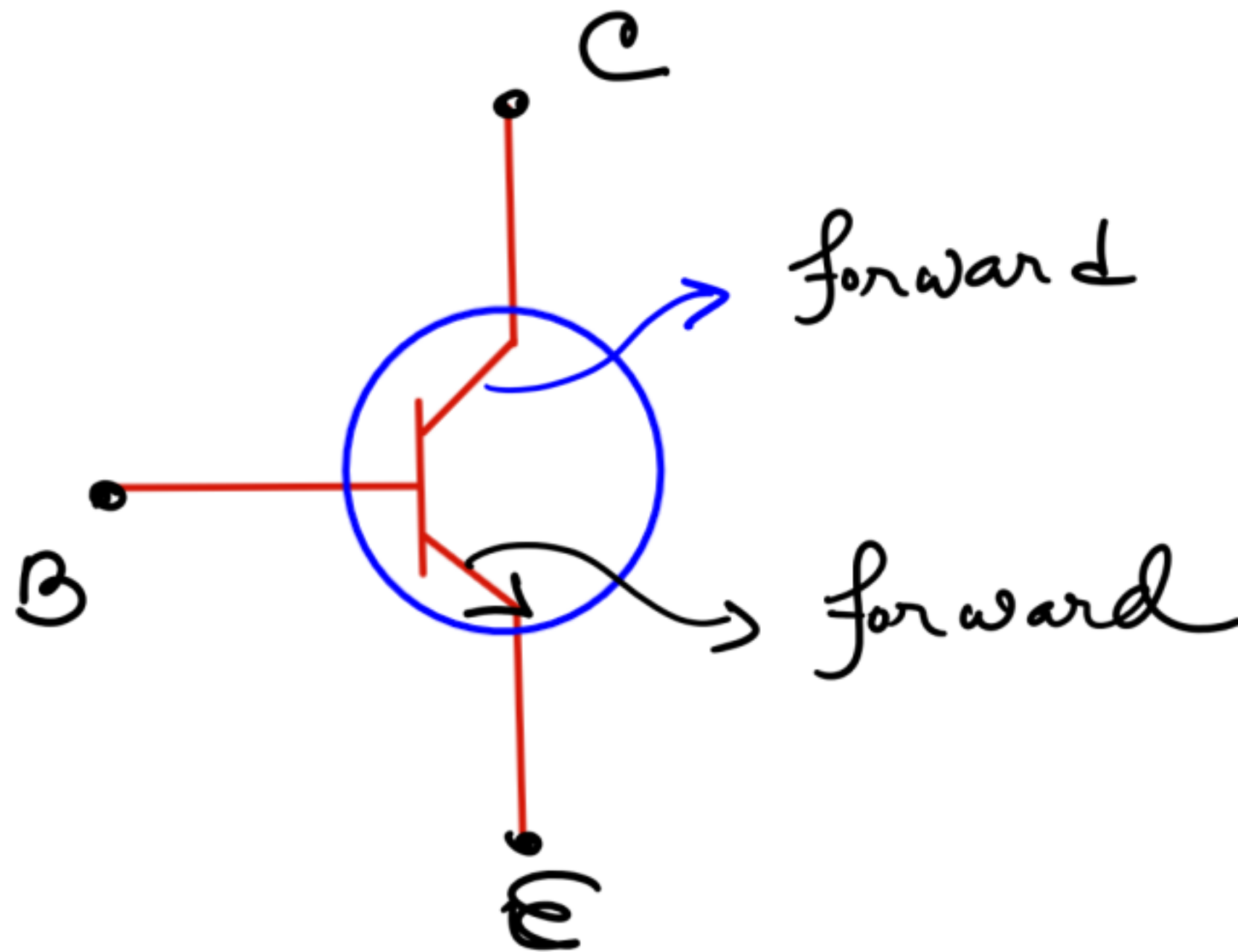
Active Mode (npn transistor)



$$V_{CE} = V_{CB} + V_{BE} > (-0.4) + 0.7 = 0.3V$$



Saturation Mode (npn transistor)

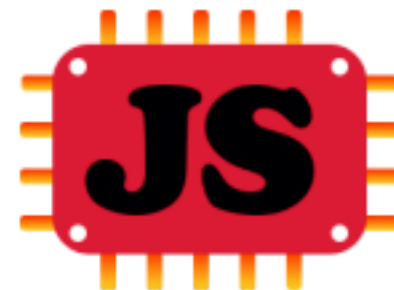


- $V_{BE} = 0.7\text{V}$

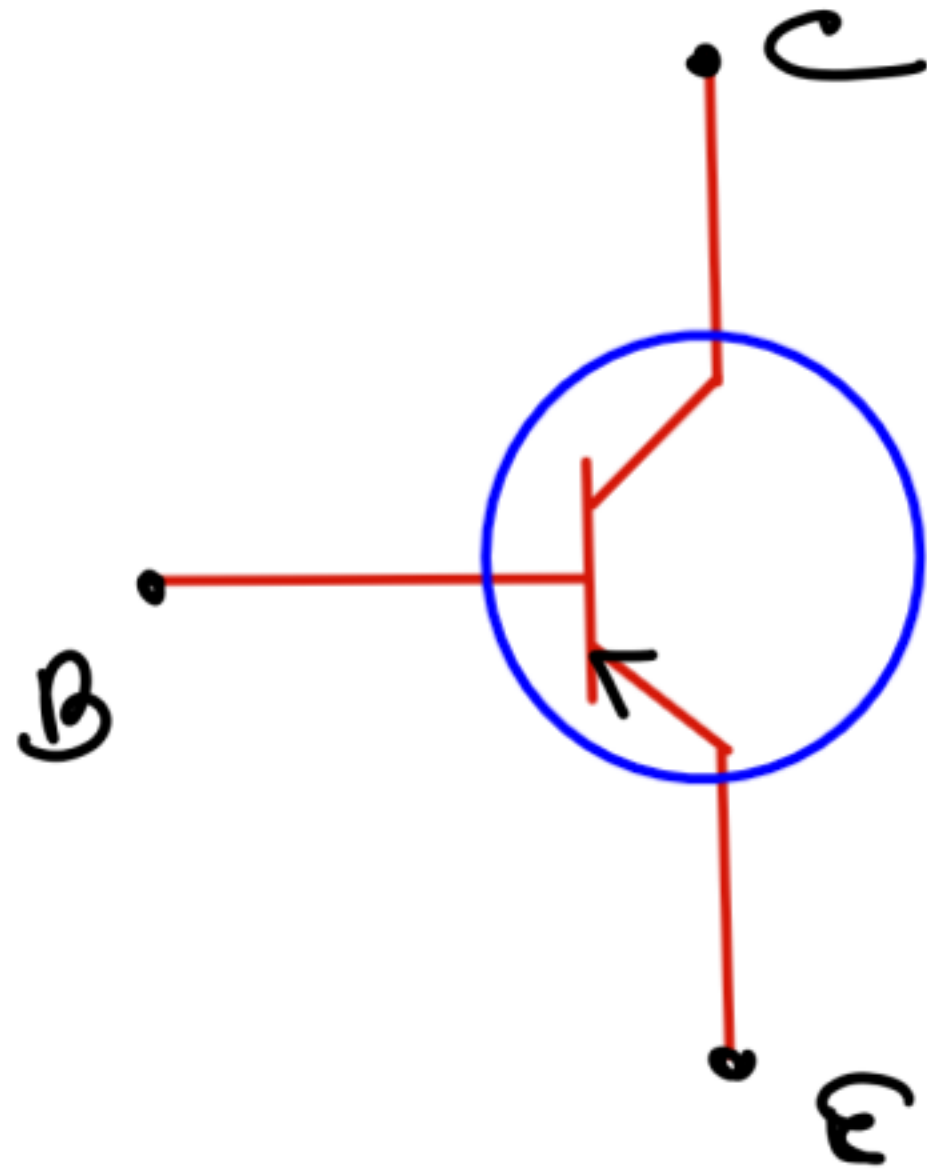
- $V_{BC} = 0.5\text{V}$

$$\therefore V_{CE} = V_{CB} + V_{BE} \\ = -0.5 + 0.7$$

- $V_{CE} = 0.2\text{V}$



Active Mode (pnp transistor)

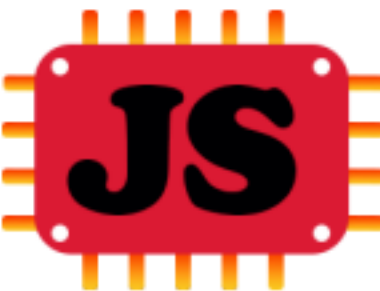
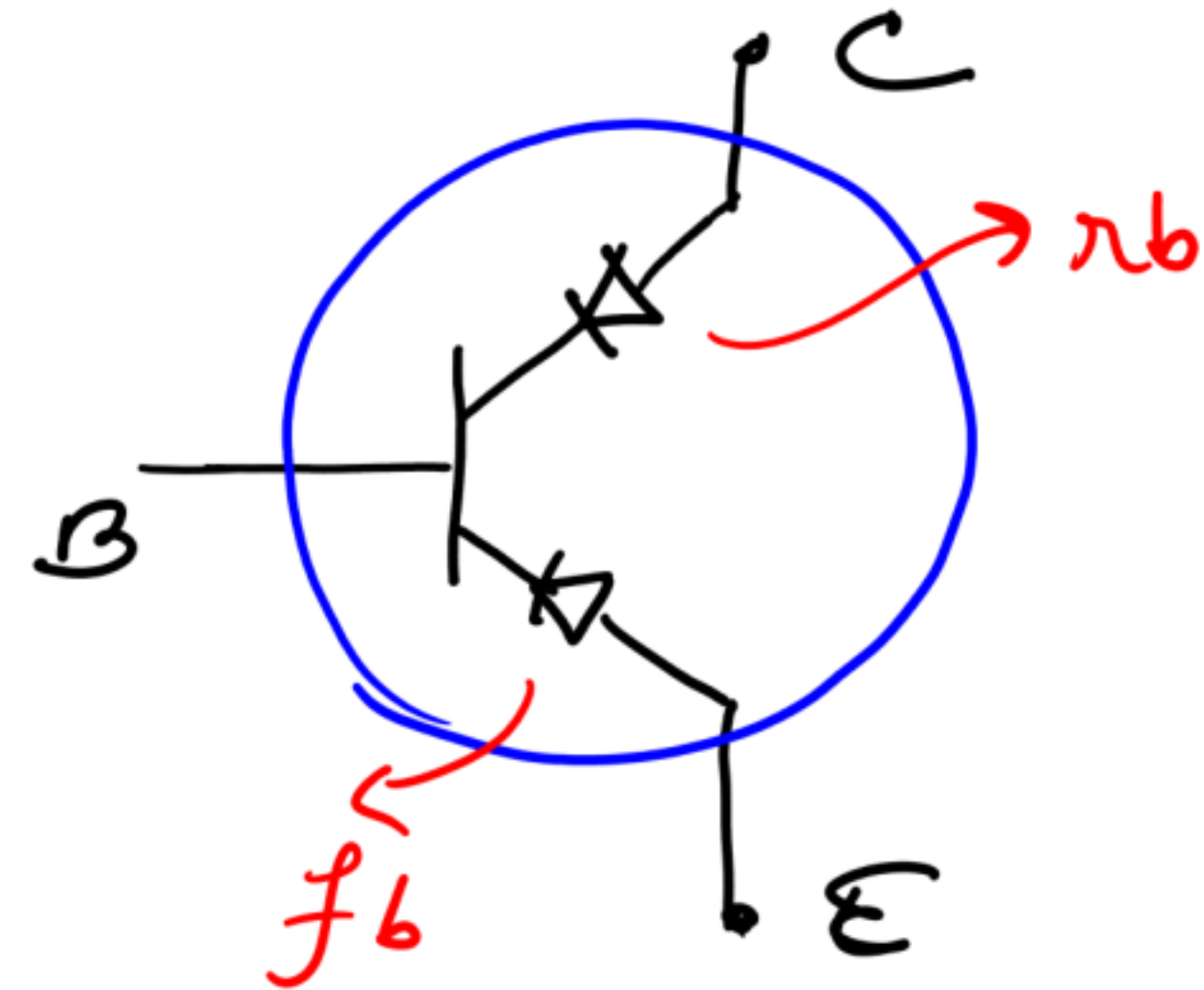
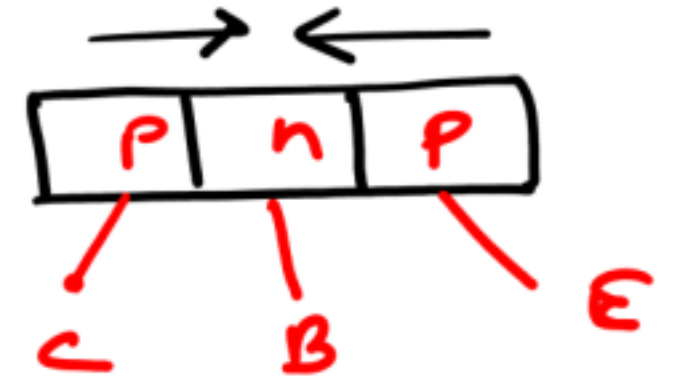


- $V_{EB} = 0.7V$

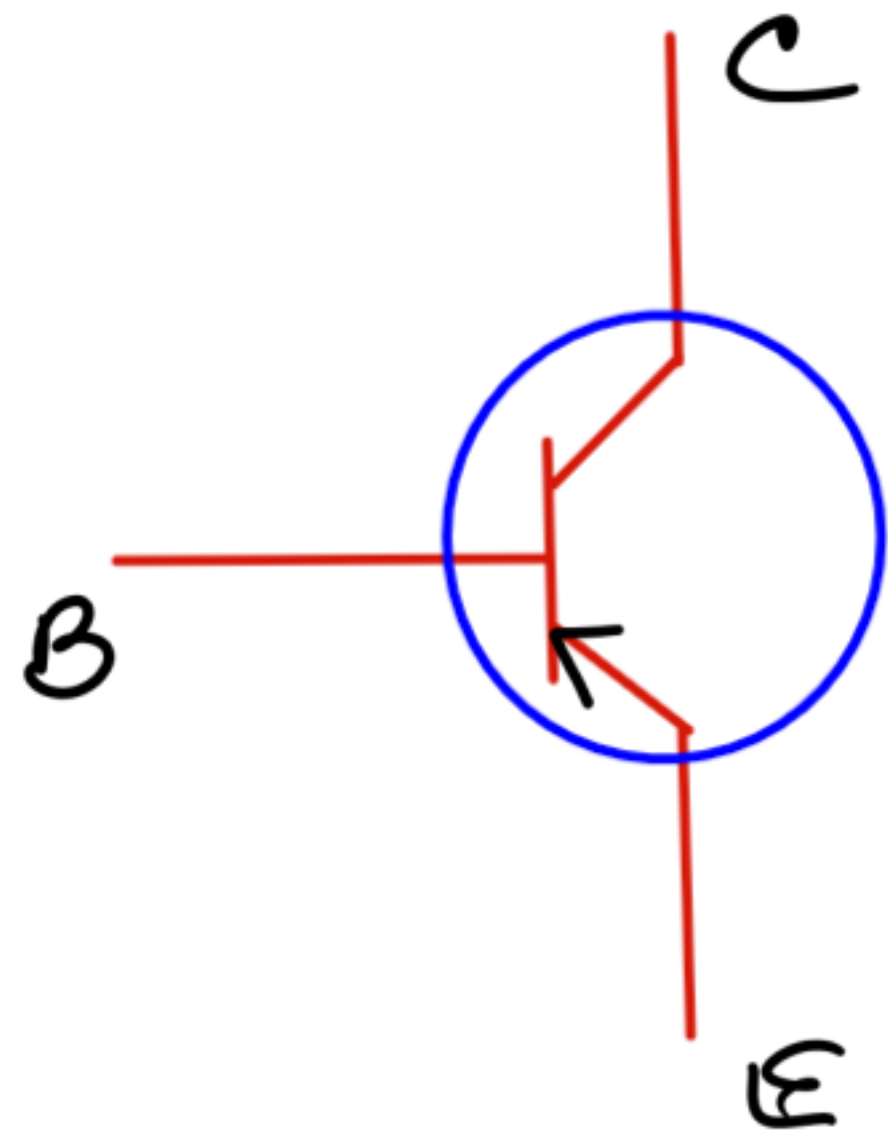
- $V_{EB} < 0.4V$

$$V_{EC} = V_{EB} + V_{BC}$$

- $V_{EC} > 0.3V$



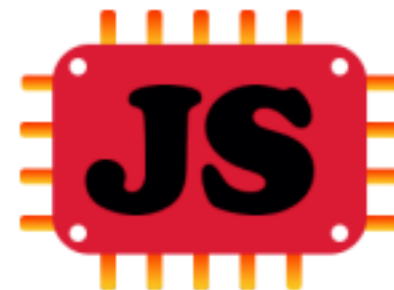
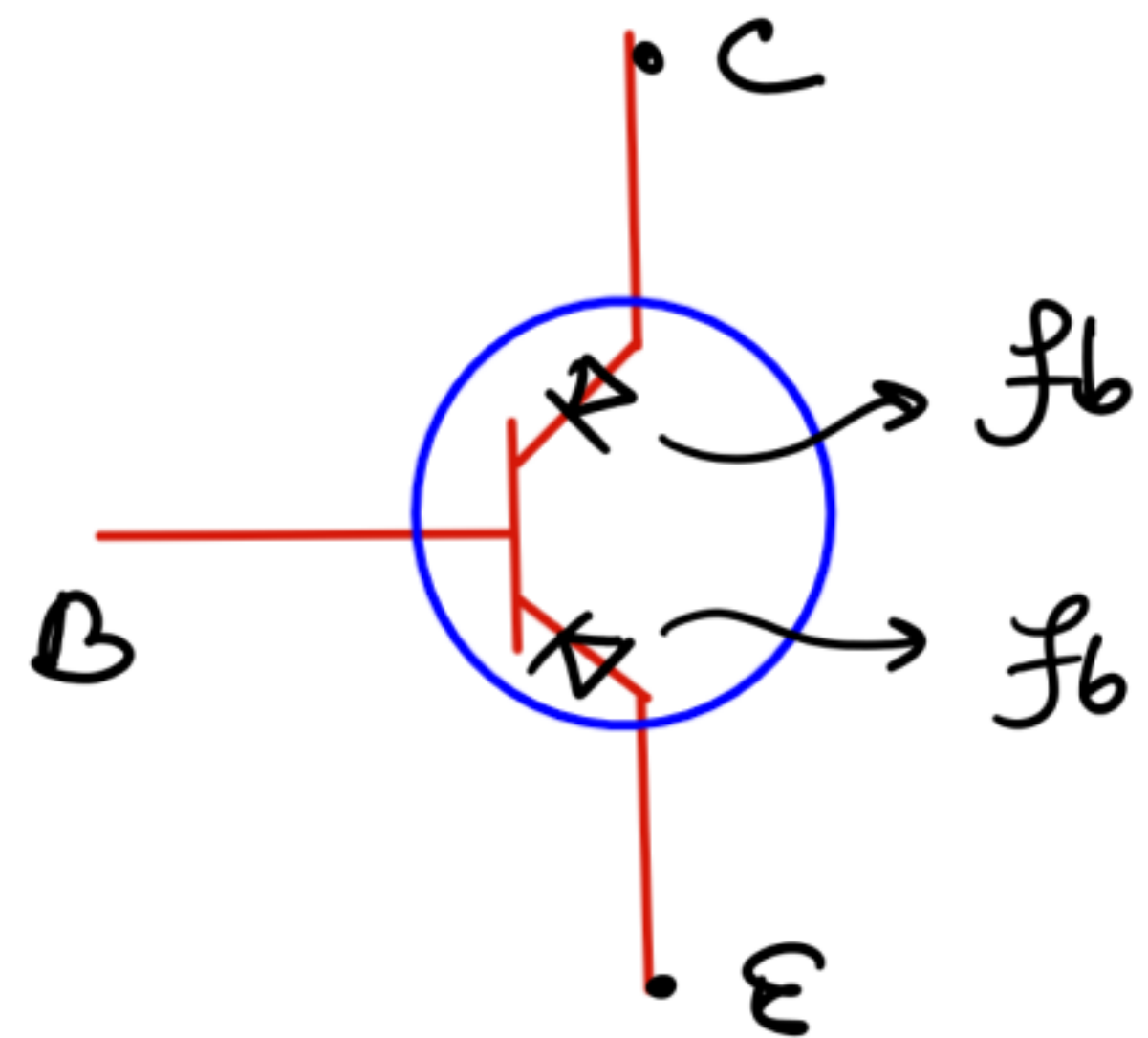
Saturation Mode (pnp transistor)



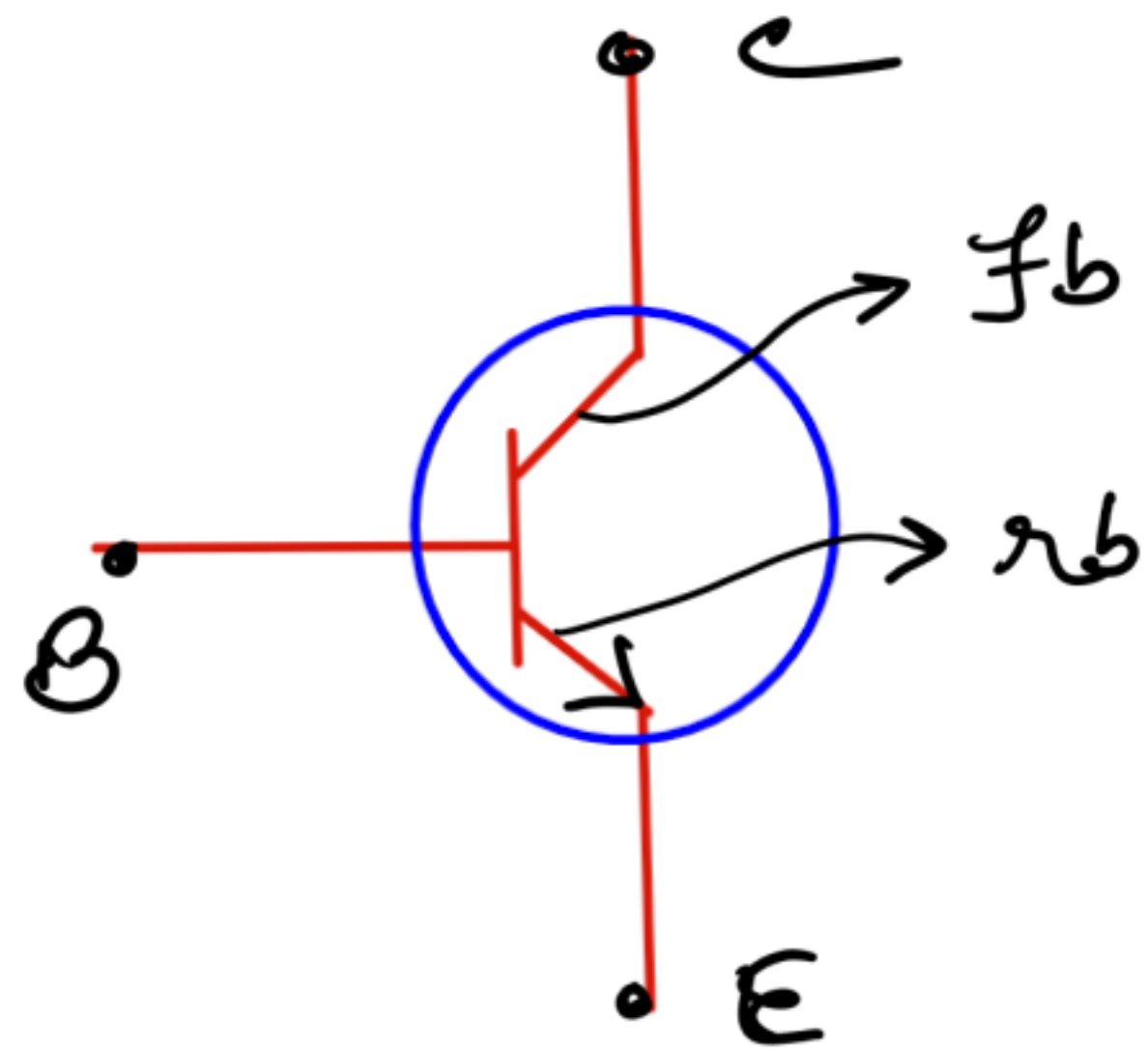
• $V_{EB} = 0.7V$

• $V_{CB} = 0.5V$

• $V_{EC} = 0.2V$



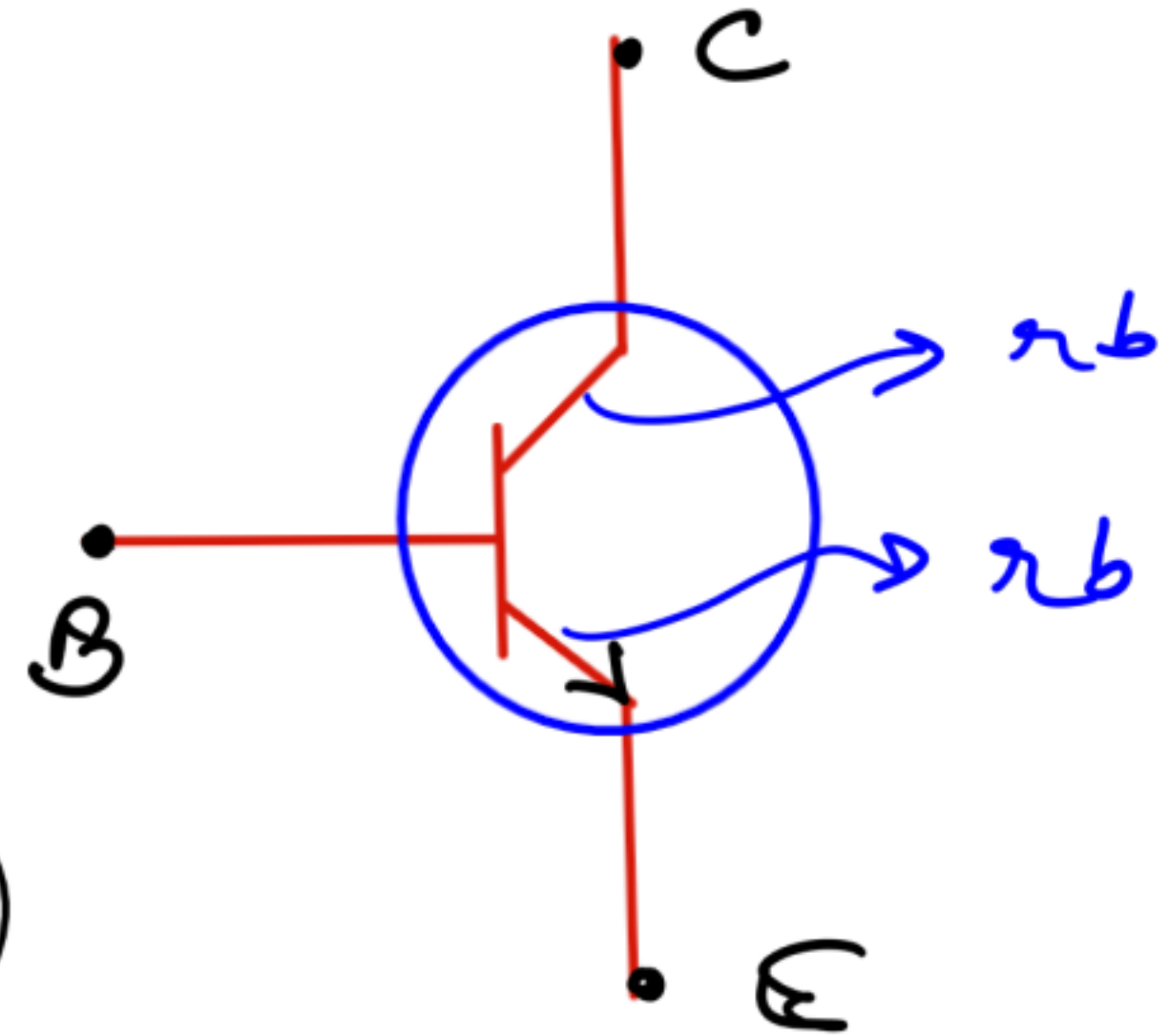
Reverse-active & Cut-off Mode



(npn)

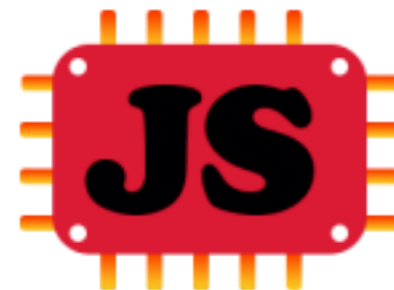
[Reverse Active]

$$V_{BE} < 0.7V$$



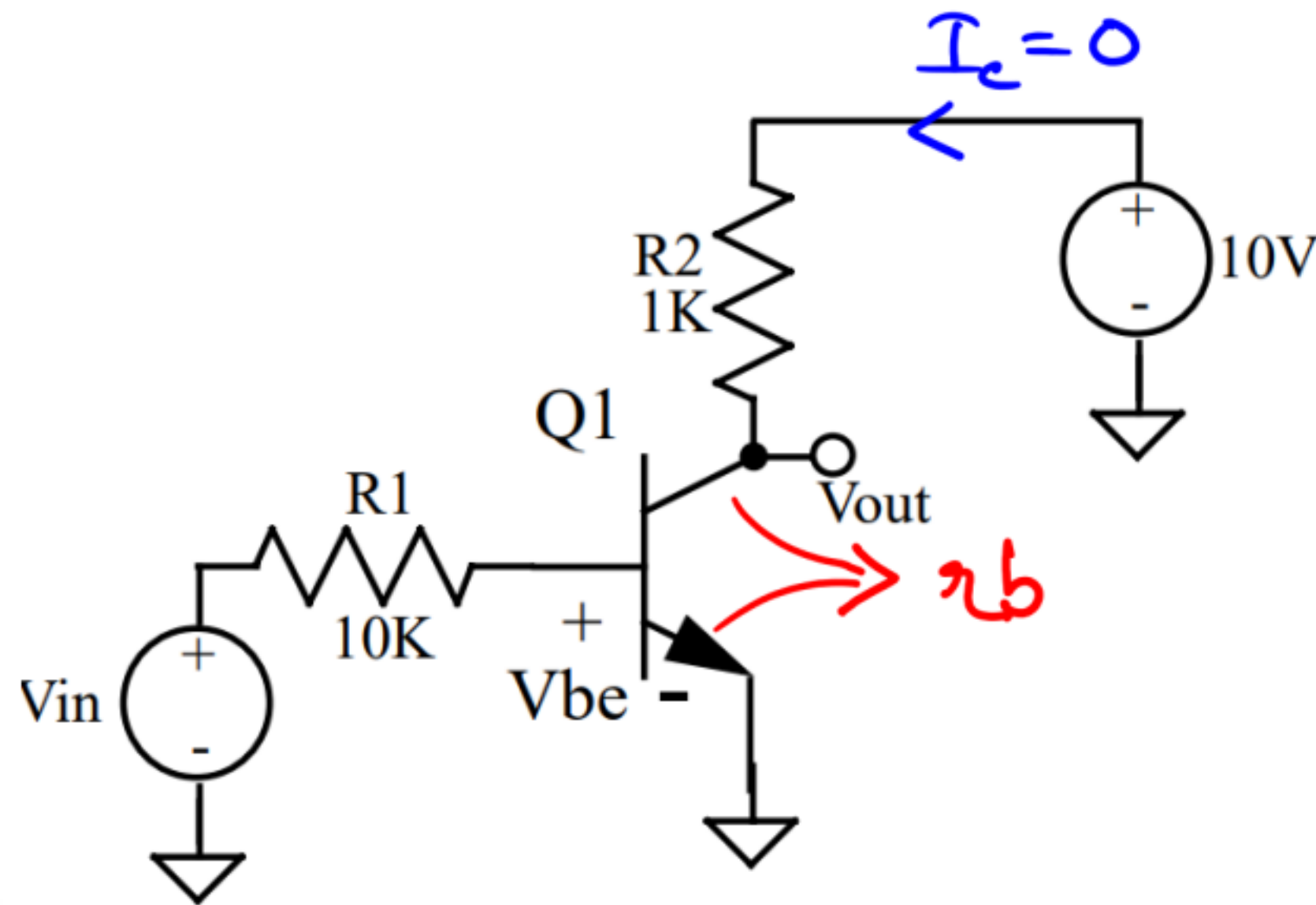
(npn)

[Cut-off]



Understanding the 3 modes of operation (1)

- Case 01: cut-off (V_{in} not enough to make B-E forward biased)

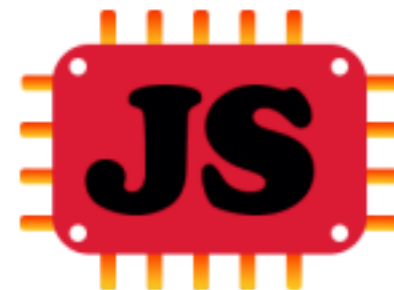


say $V_{in} = 0.2V < 0.7V$

$\therefore$ Cut-off

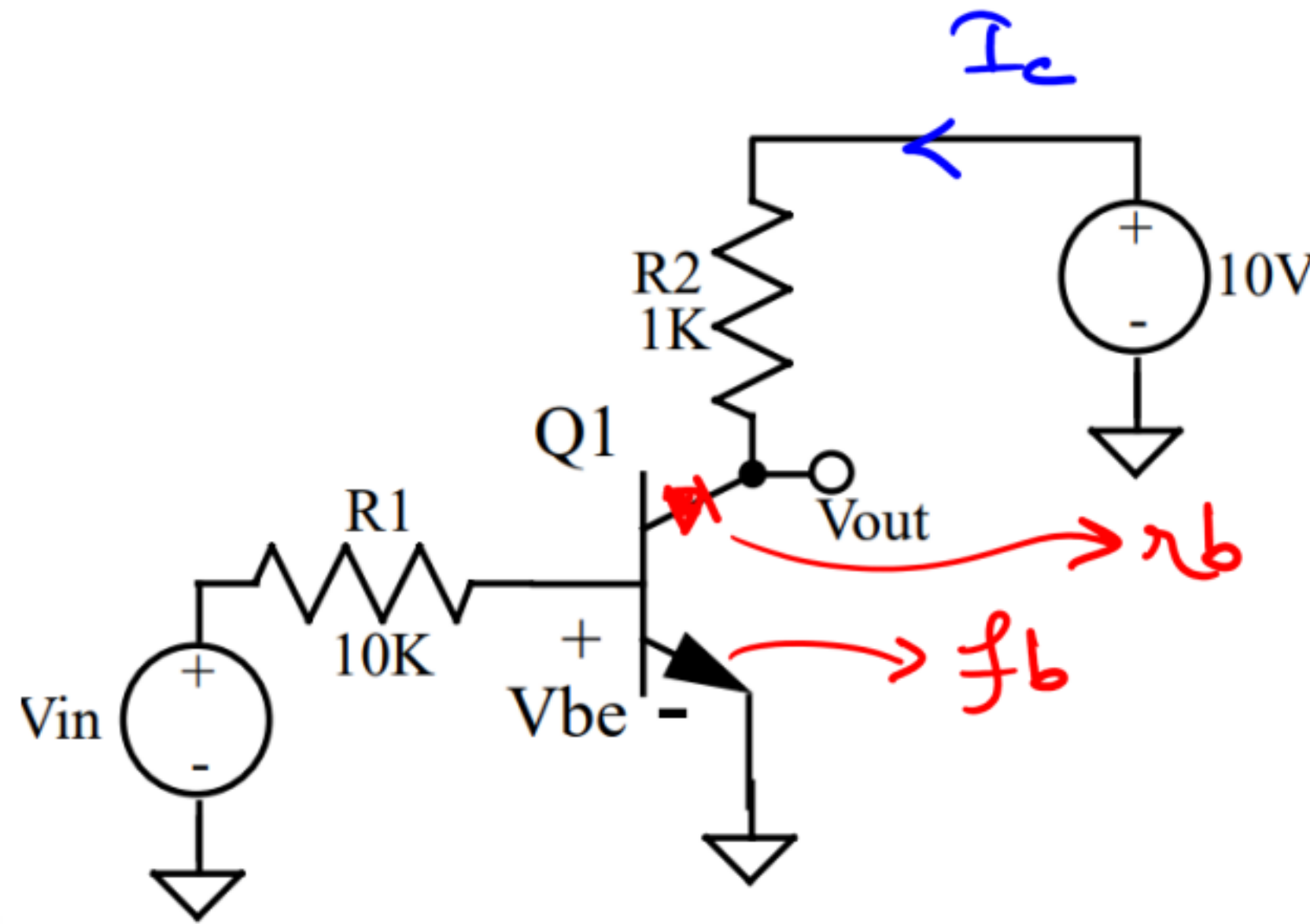
$$I_B, I_C, I_E = 0$$

$$\therefore V_{out} = 10V$$



Understanding the 3 modes of operation (2)

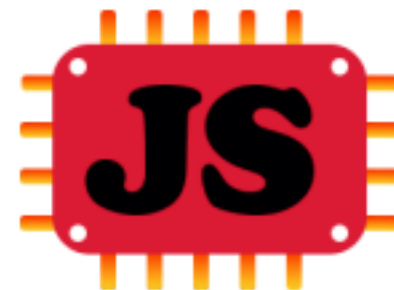
- Case 02: Active (V_{in} enough to make B-E forward biased, but B-C reverse biased)



say, $V_{in} = 1V > 0.7V$

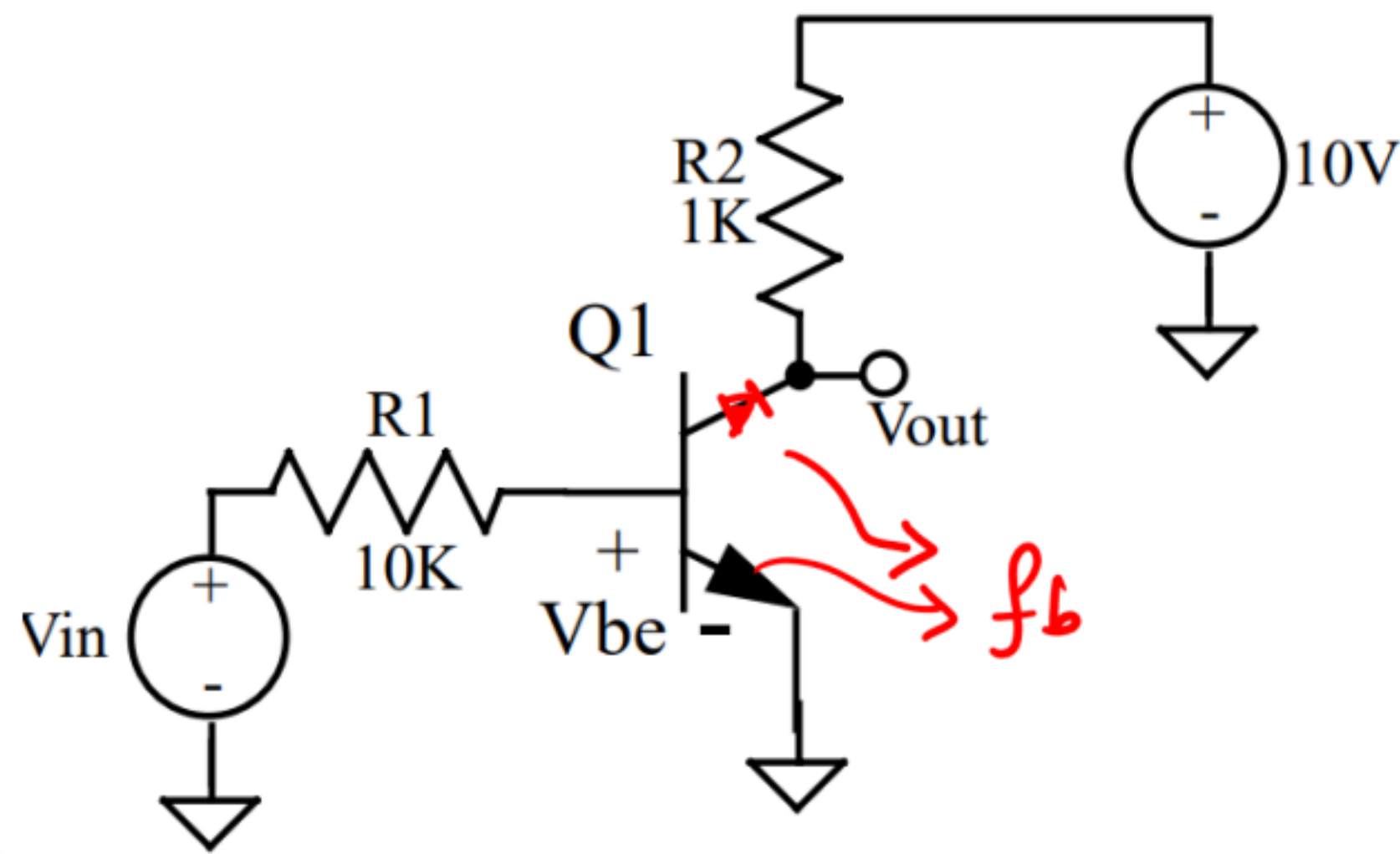
$\therefore$ active

$\therefore V_{out} = 10 - I_c R_2 < 10$



Understanding the 3 modes of operation (3)

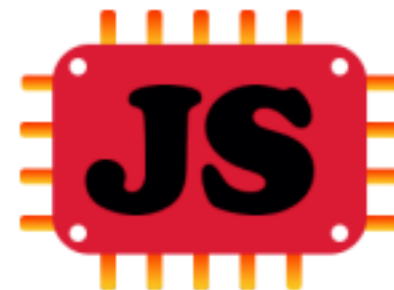
- Case 03: saturation (V_{in} enough to make both B-E & B-C forward biased)



$$V_{in} = 15V \text{ (say)}$$

$\therefore$ saturation

$$\therefore V_{out} = V_{CE} \approx 0.2V$$



Characteristic curves for BJT (1)

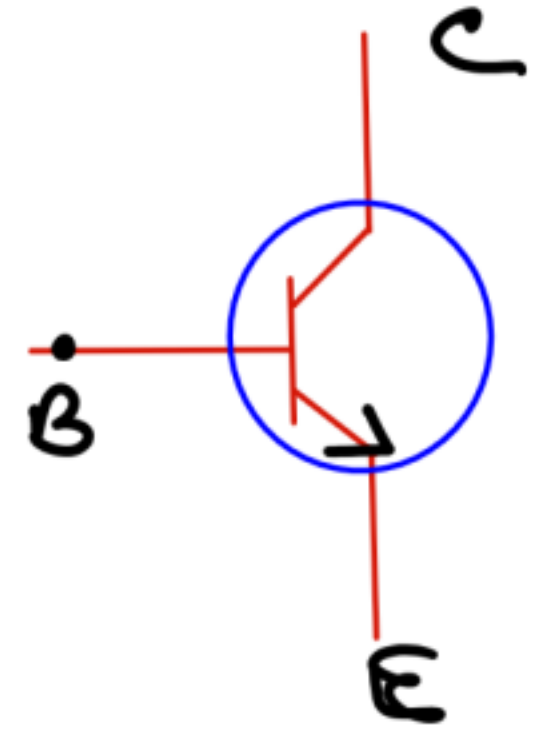
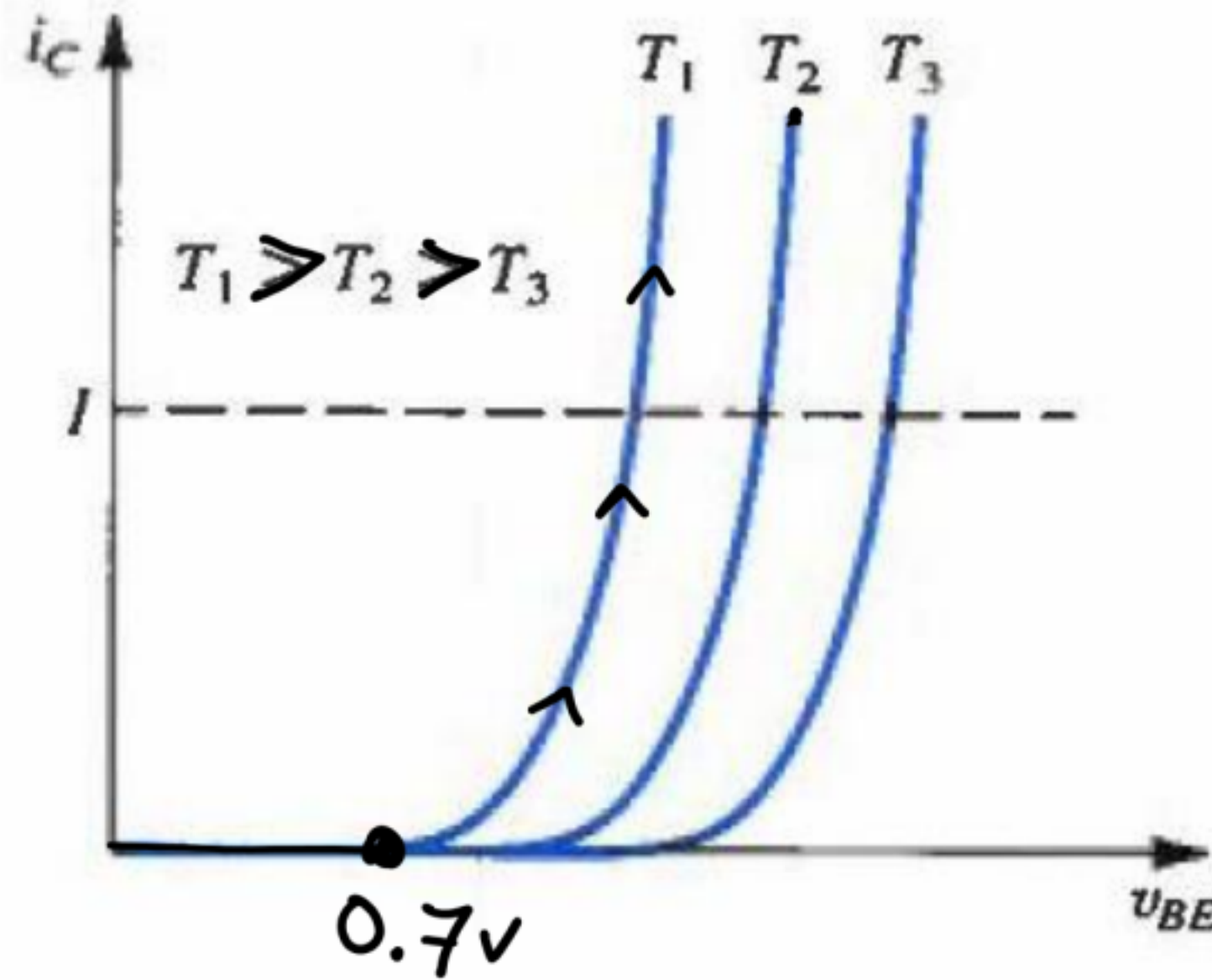
$$i_c = i_s e^{\frac{V_{BE}}{V_T}} \quad \text{--- ①}$$

✗

$$I_c \leftrightarrow V_{BE}$$

$$I_c \leftrightarrow V_{CB}$$

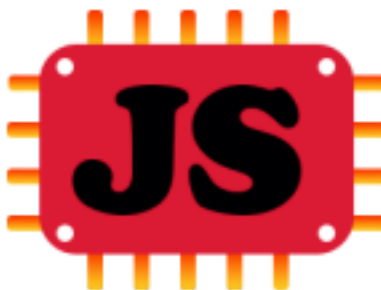
$$I_c \leftrightarrow V_{CE}$$



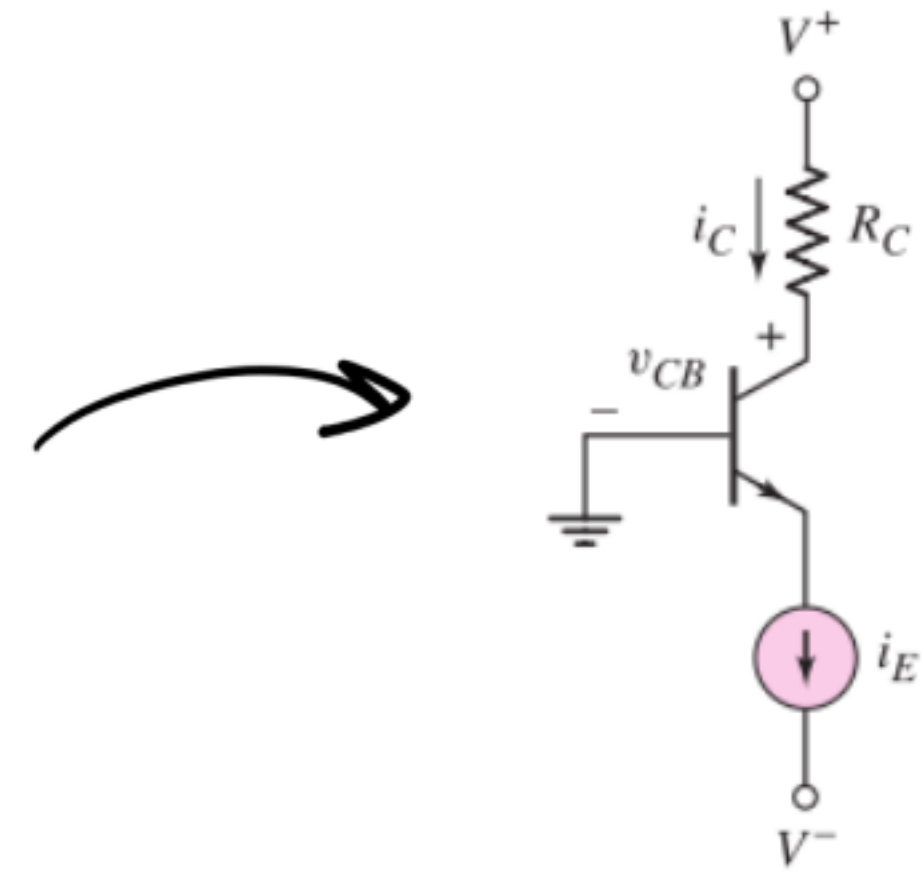
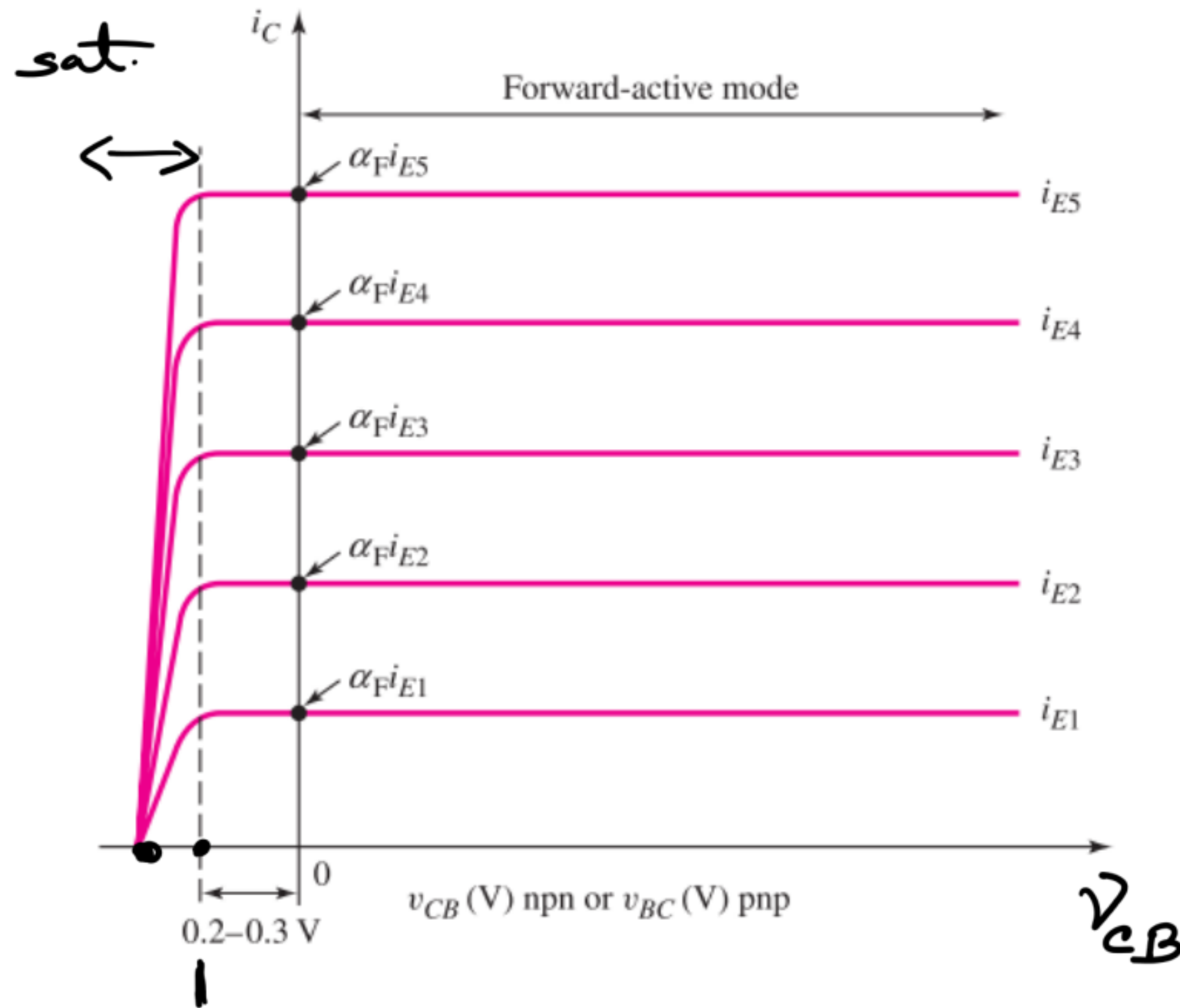
$V_T \rightarrow$ thermal voltage

$$V_T = \frac{KT}{q}$$

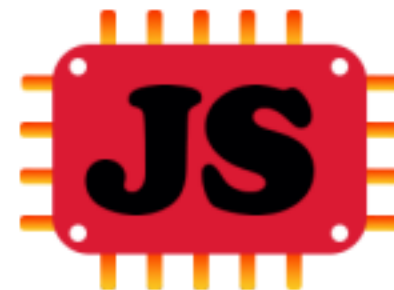
$$K = \frac{IR}{N_A}$$



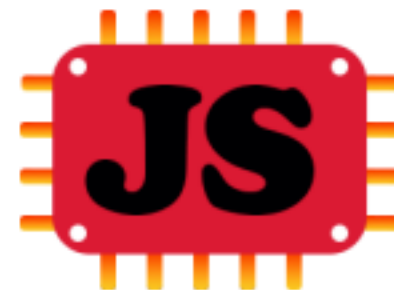
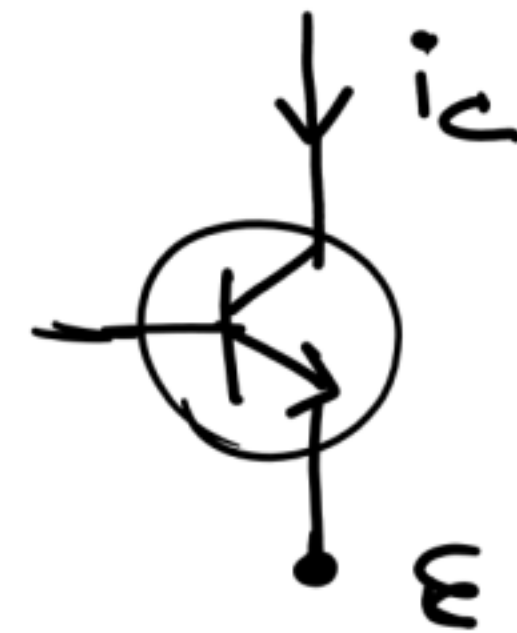
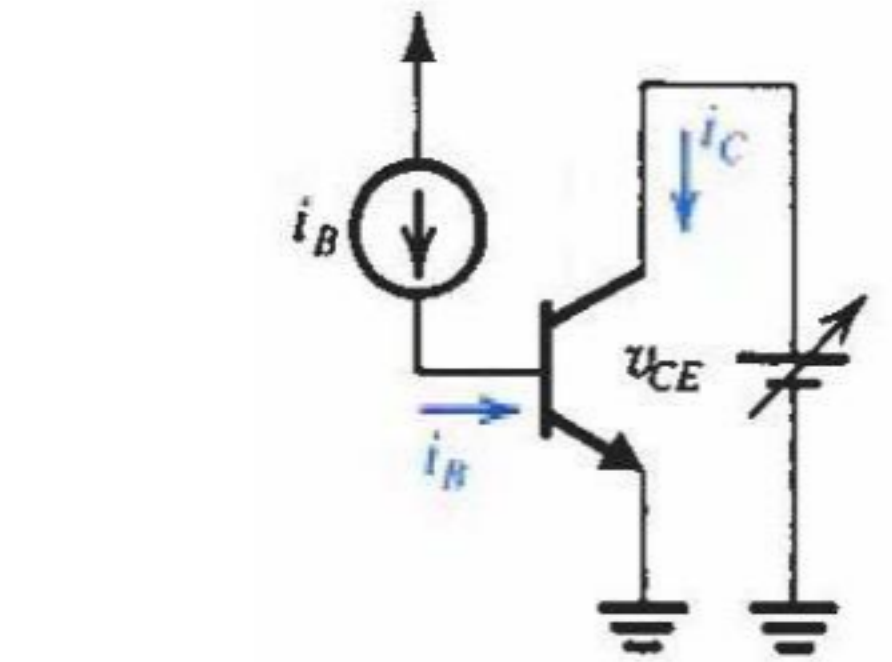
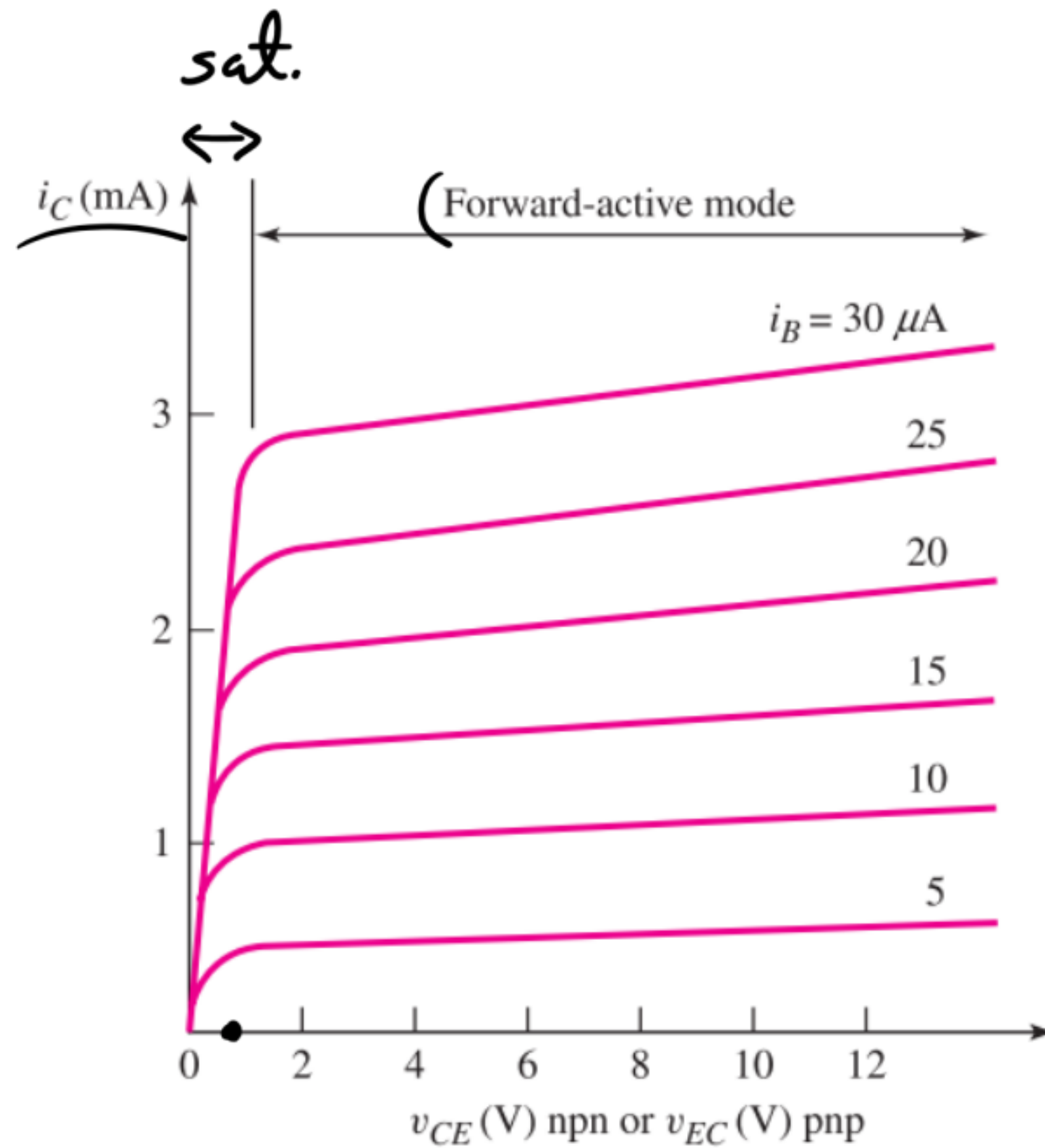
Characteristic curves for BJT (2)



$$v_{EB} > -0.4 \text{ V}$$



Characteristic curves for BJT (3)



Current Gains α, β

- Applicable for active mode

- Common-base current gain, $\alpha = \frac{I_c}{I_E}$

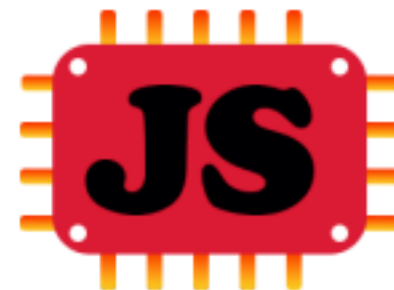
- Common-emitter current gain, $\beta = \frac{I_c}{I_B}$

- Relationship between α and β >>

$$\beta = \frac{\alpha}{1 - \alpha}$$

$$\alpha = \frac{\beta}{1 + \beta}$$

$$\begin{aligned}\alpha &= \frac{I_c}{I_E} = \frac{I_c}{I_c + I_B} \\ &= \frac{I_c/I_B}{I_c/I_B + 1} \\ \alpha &= \frac{\beta}{1 + \beta}\end{aligned}$$

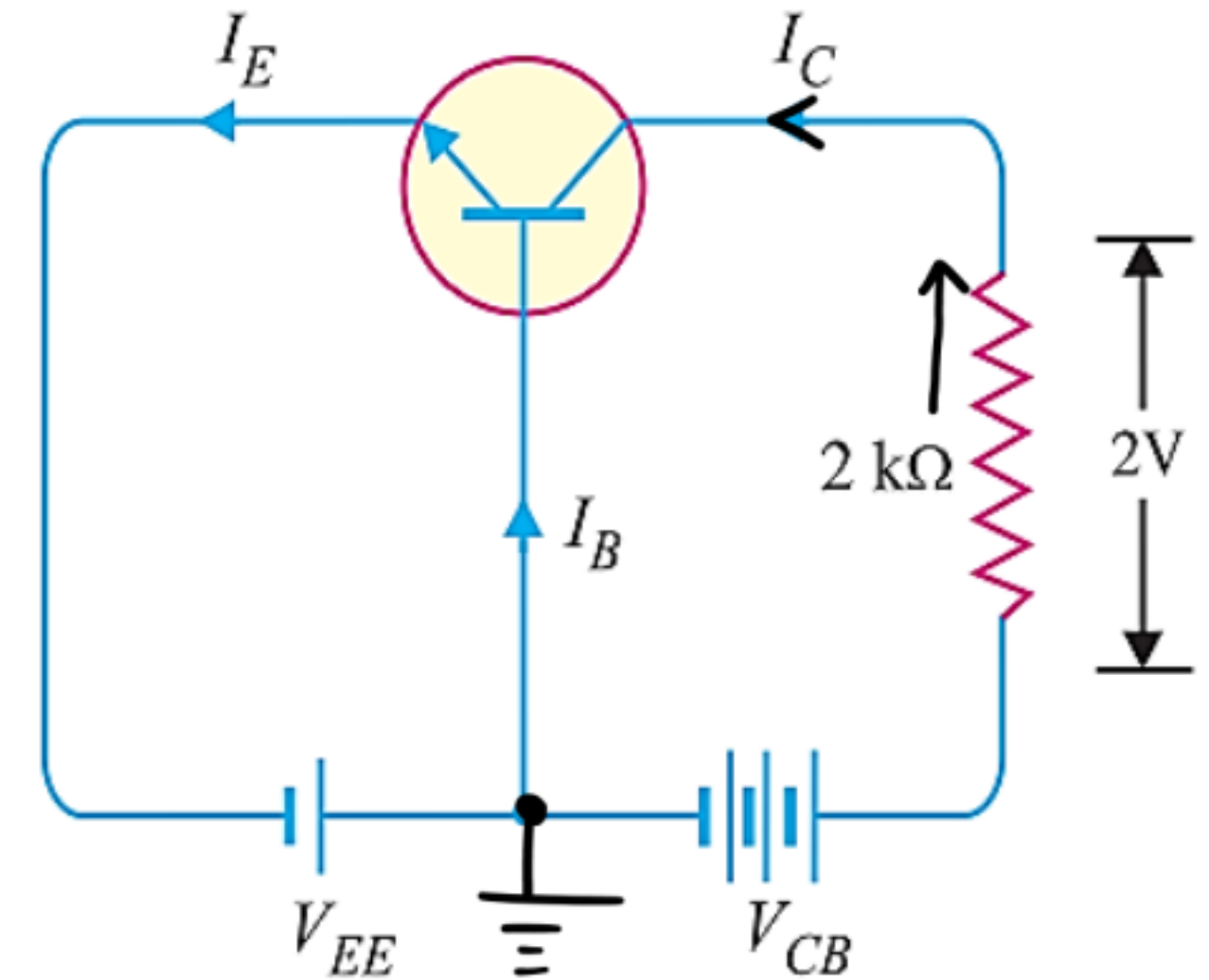


Example 8.6. In a common base connection, $\alpha = 0.95$. The voltage drop across $2\text{ k}\Omega$ resistance which is connected in the collector is 2 V . Find the base current.

$$\hat{I}_c = \frac{2}{2\text{ k}} = 1\text{ mA}$$

$$\alpha = \frac{I_c}{I_E} \quad \therefore \hat{I}_E = \frac{1\text{ mA}}{0.95} = 1.05\text{ mA}$$

$$\therefore \hat{I}_b = I_E - I_c = 0.05\text{ mA}$$



Edge of Saturation

- Edge of saturation or soft saturation is just a region where the V_{ce} nearly saturates but the current is still affected by I_B (base current)

- $Overdrive\ Factor\ (OF) = \frac{I_B(saturation)}{I_B(edge\ of\ saturation)} = \frac{I_B(sat)}{I_B(EOS)}$

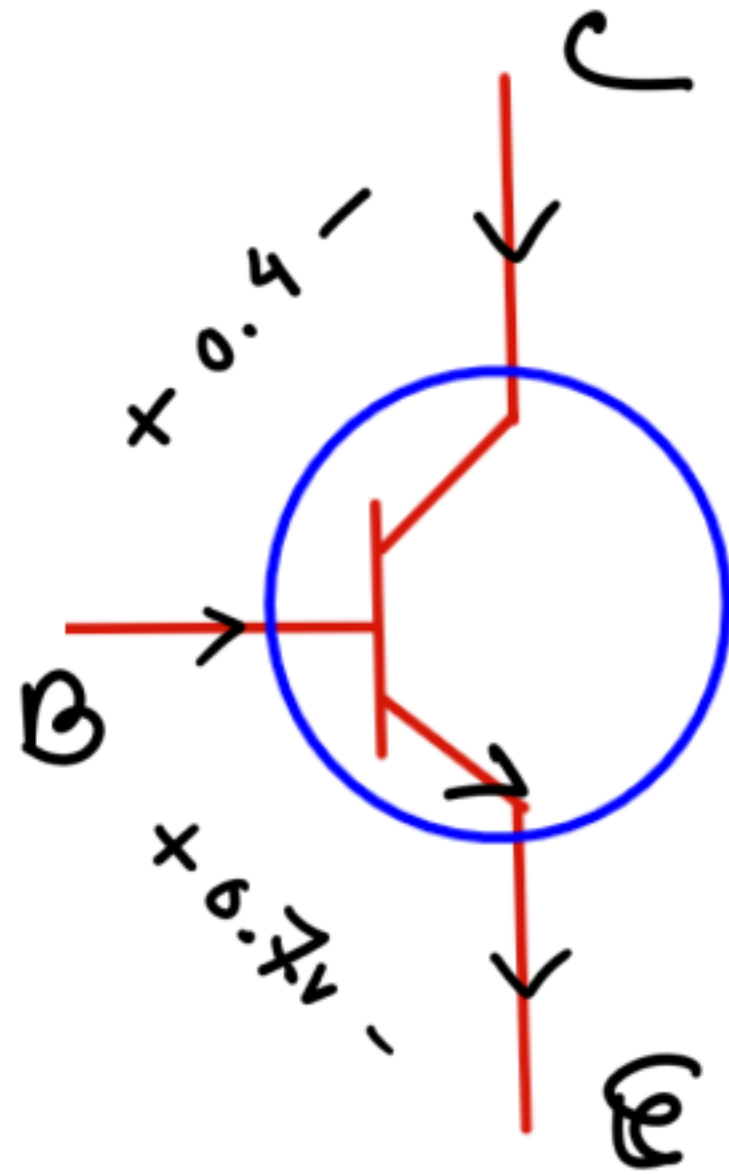
- $\beta_{min} = \frac{I_C(saturation)}{I_B(EOS)}$

$$OF = ratio = \frac{I_B(sat)}{I_B(eos)}$$

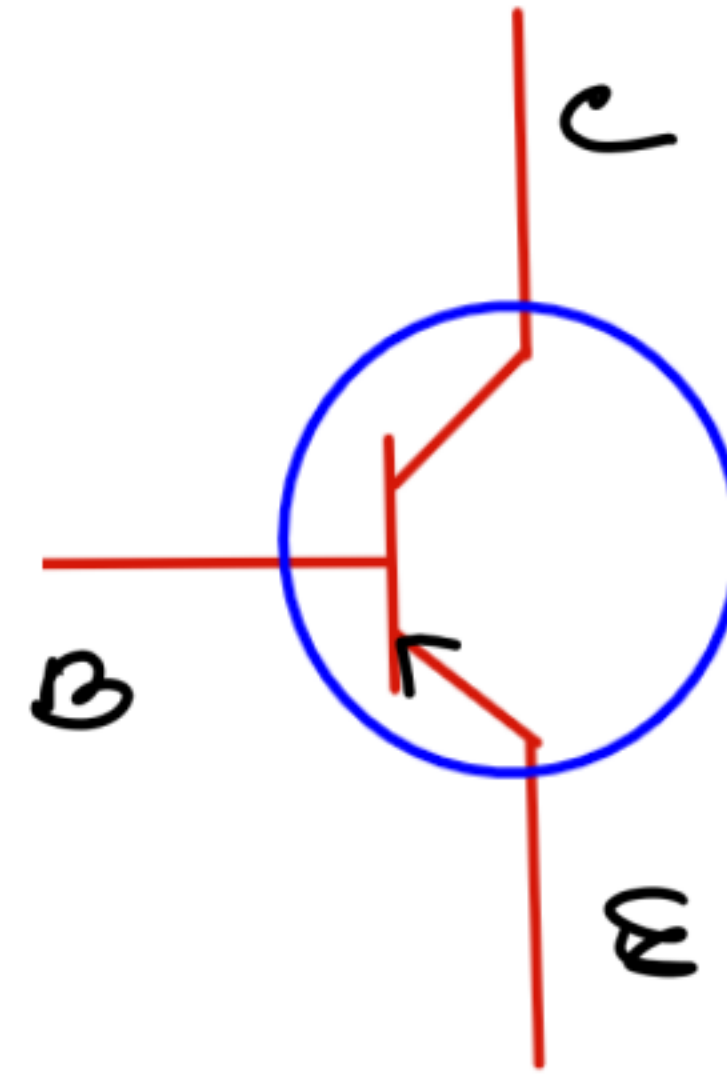
β_{De} β_{min} β_{forced}



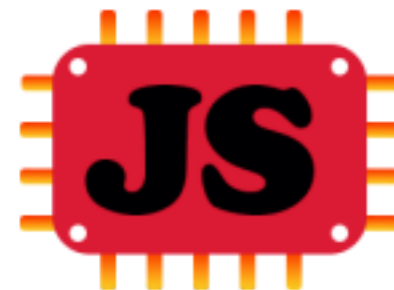
Junction voltages at Edge of Saturation condition



$$\begin{aligned}V_{BE} &= 0.7V \\V_{CE} &= 0.3V \\V_{BC} &= 0.4V\end{aligned}$$

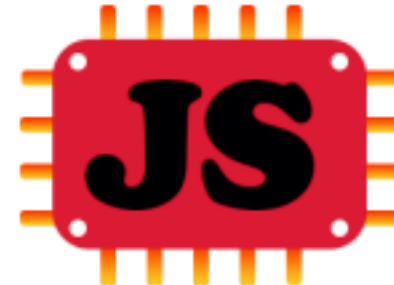


$$\begin{aligned}V_{CE} &= 0.4V \\V_{BE} &= 0.7V \\ \therefore V_{EC} &= 0.3V\end{aligned}$$



Current gain, β in saturation mode

- In saturation mode, as V_{BC} increases, I_B increases, I_C decreases.
- We can control β using V_{BC}
- $$\beta_{forced} = \frac{I_C(saturation)}{I_B(saturation)}$$
- We can force the β to be lower in the saturation mode than in the active mode.
- $\beta_{forced} < \beta$ (active mode) — 



Determine whether or not the transistor in Figure 4-16 is in saturation. Assume $V_{CE(sat)} = 0.2 \text{ V}$.

Let, transistor is in saturation mode

$$V_{CE} = V_C - V_E \quad \therefore V_C = V_{CE} + V_E$$

$$= 0.2 + 0$$

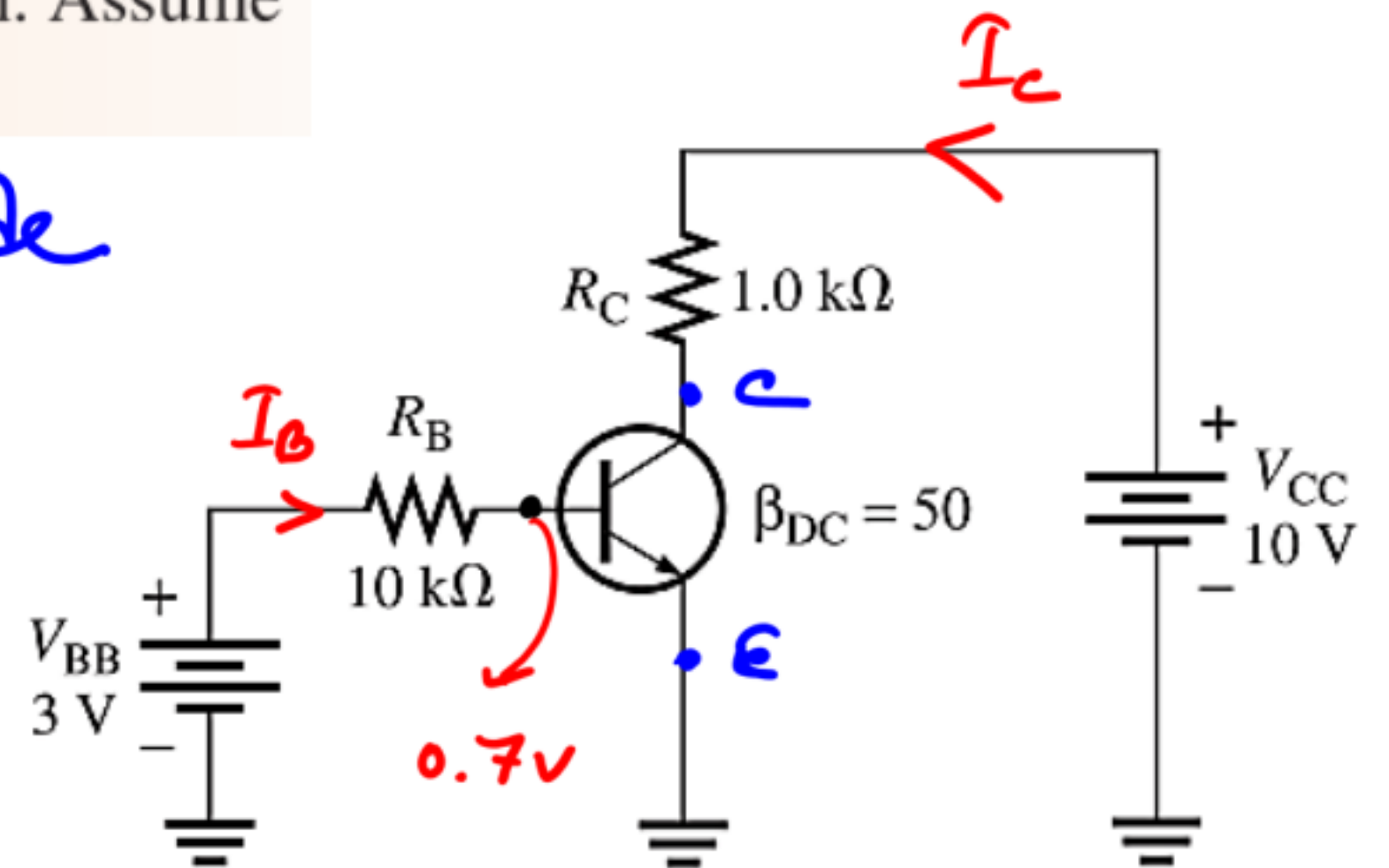
$$V_{BE} = 0.7$$

$$= 0.2 \text{ V}$$

$$\therefore V_B = 0.7 + V_E = 0.7 \text{ V}$$

$$\therefore I_B(sat) = \frac{V_{BB} - V_B}{R_B} = \frac{3 - 0.7}{10 \text{ k}} = 0.23 \text{ mA}$$

$$\therefore I_C(sat) = \frac{V_{CC} - V_C}{R_C} = 9.8 \text{ mA}$$



$$\beta_{\text{forced}} = \frac{I_C(sat)}{I_B(sat)}$$

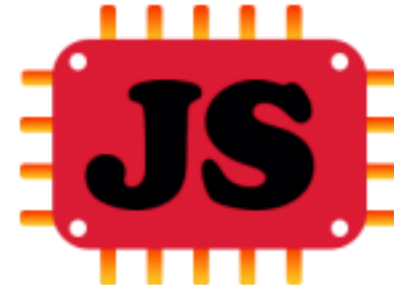
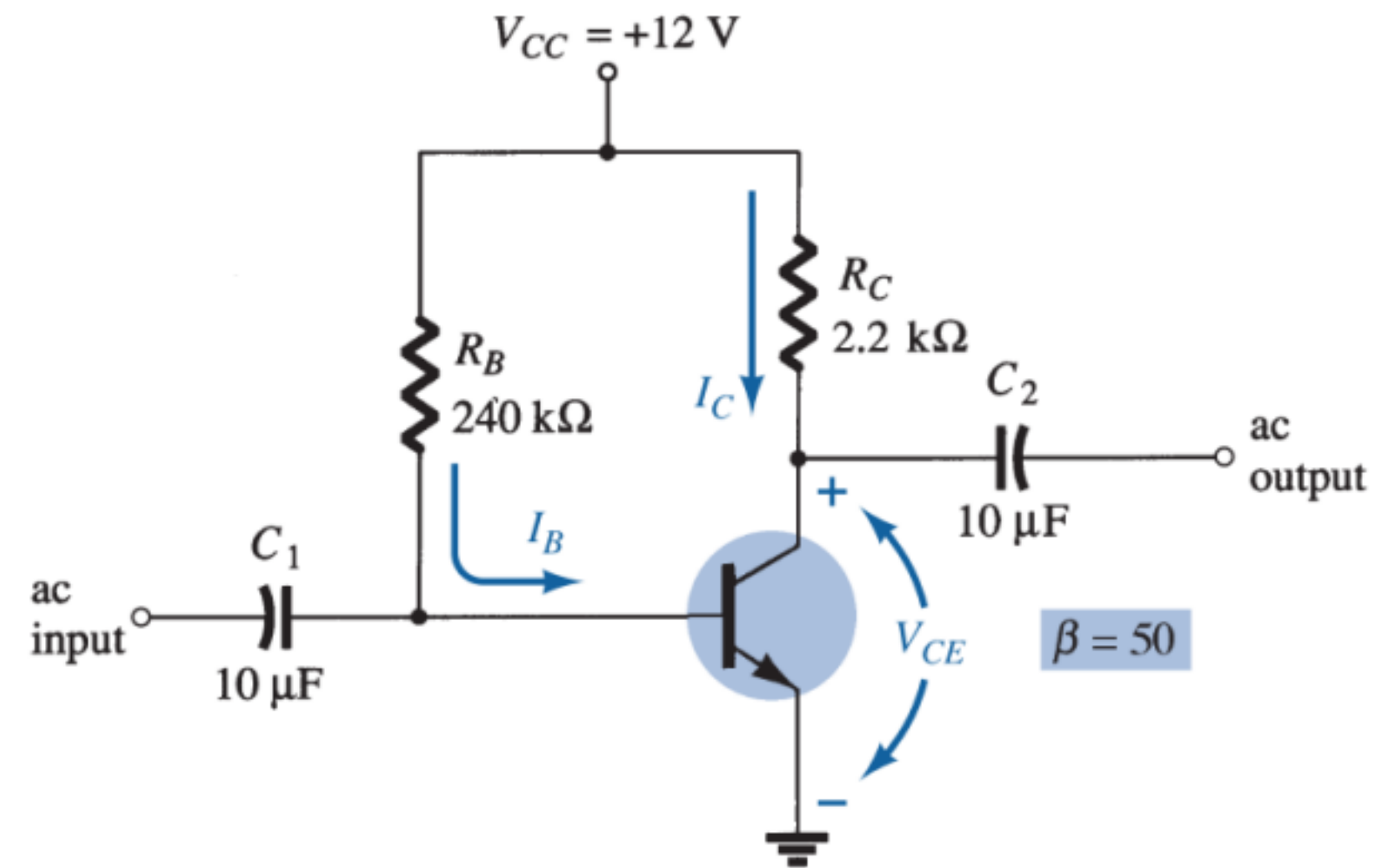
$$\therefore \beta_f = \frac{9.8}{0.23} = 42.6 < \beta_{DC}$$

$\therefore$ Yes, transistor in saturation



DC Biasing of BJT

- For amplification of AC voltages, the BJT must be made on and it should be working in the **linear region (active mode)**
- We apply DC voltages for this very reason!



Determine the following for the fixed-bias configuration of Fig. 7.

- a. I_{BQ} and I_{CQ} .
 b. V_{CEQ} .
 c. V_B and V_C .
 d. V_{BC} .

Let $\rightarrow$ active mode

$$\therefore V_{BE} = 0.7 \rightarrow V_B = 0.7V$$

$$I_{BQ} = \frac{12 - 0.7}{240K} = 0.047mA$$

$$I_{CQ} = \beta I_{BQ} = 2.35mA$$

$$\therefore V_C = V_{CC} - I_C R_C = 12 - (2.35)(2.2) = 6.82V$$

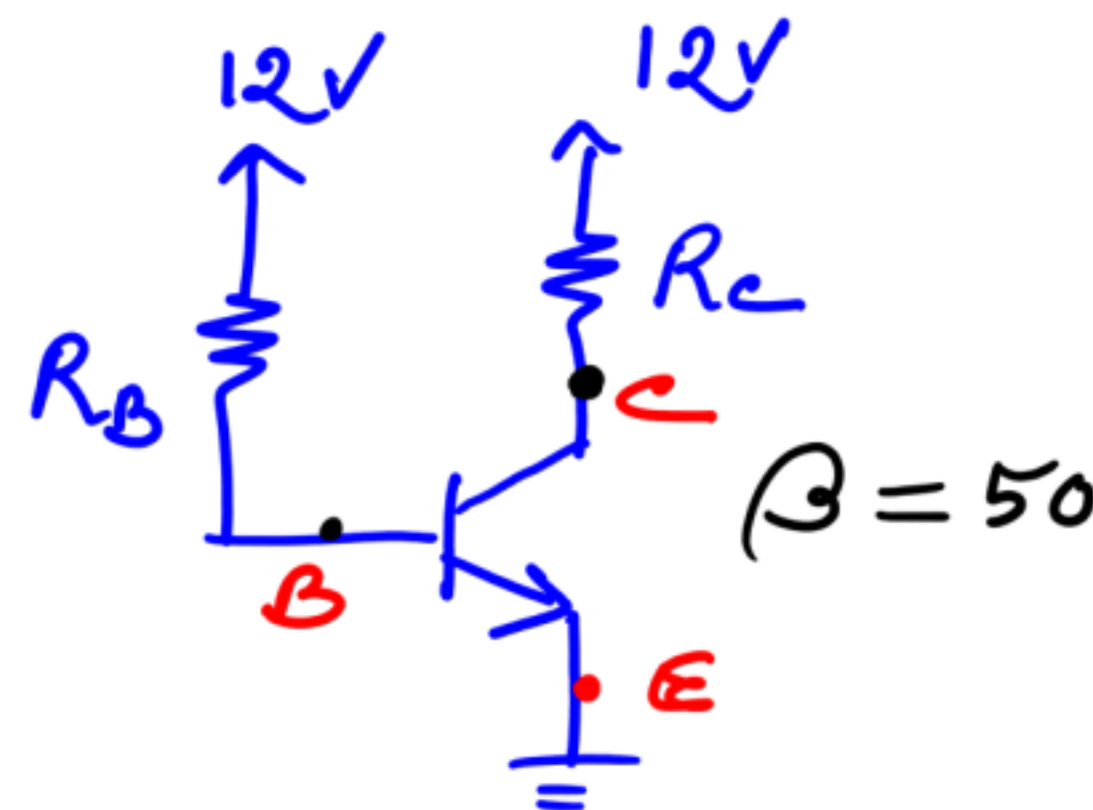
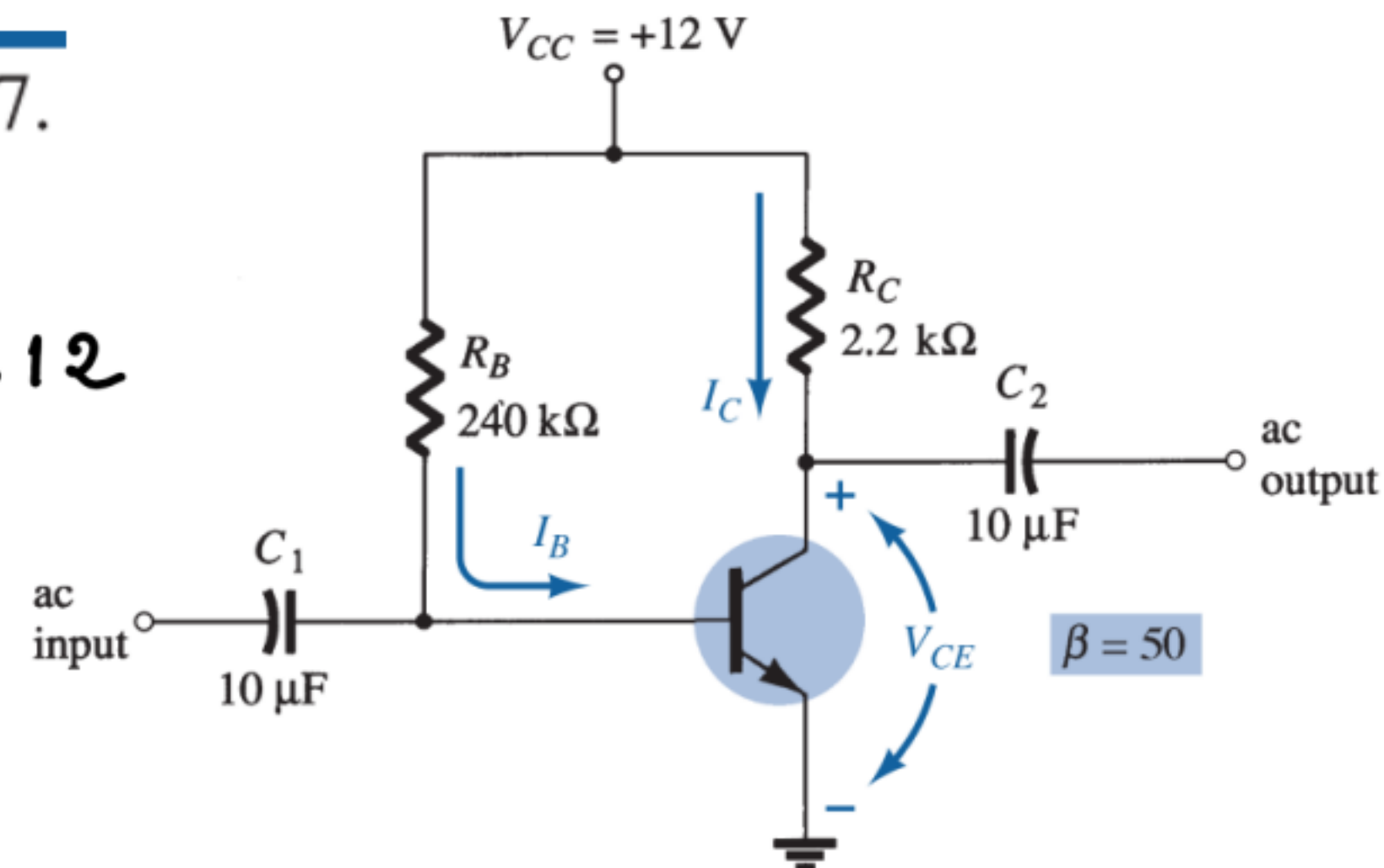
$$V_{CE} = V_C$$

Check:

$$V_{BC} = V_B - V_C = -6.12$$

$$\therefore V_{BC} < 0.4V$$

So; *active true*



BJT DC Circuit Analysis



Determine I_B , I_C , I_E , V_{BE} , V_{CE} , and V_{CB} in the circuit of Figure 4-9. The transistor has a $\beta_{DC} = 150$.

$$I_B = \frac{V_{BB} - V_{BE}}{R_B} = \frac{5 - 0.7}{10k} = 0.43 \text{ mA}$$
$$\beta = 150 = \frac{I_C}{I_B} \quad \therefore I_C = \beta I_B = 64.5 \text{ mA}$$

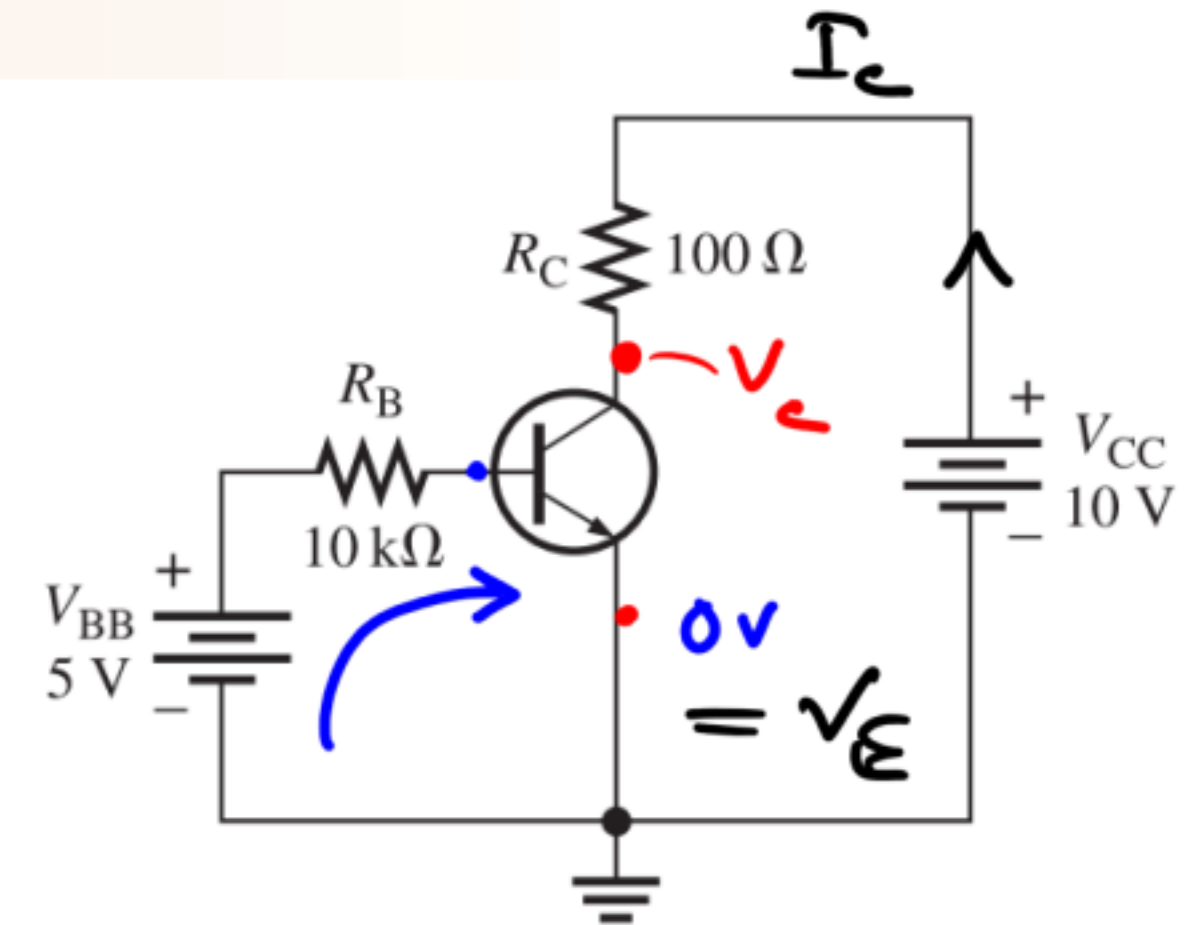
$$I_E = I_C + I_B = 64.93 \text{ mA}$$

$$V_{BE} = 0.7 \text{ V}$$

$$V_{CE} = V_C - V_E = 3.55 \text{ V}$$

$$\therefore V_C = V_{CC} - I_C R_C = 3.55 \text{ V}$$

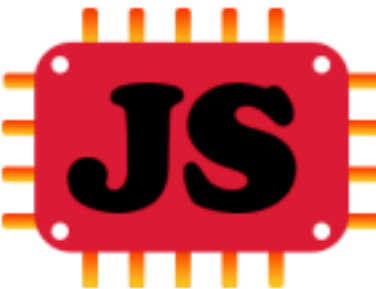
$$V_{CB} = V_C - V_B = 2.85 \text{ V}$$



$$V_{BE} = 0.7 \text{ V}$$

$$\therefore V_B = 0.7 \text{ V}$$

$$(V_{BC} = -2.85)$$



The transistor in the circuit of Fig. 6.15(a) has $\beta = 100$ and exhibits a v_{BE} of 0.7 V at $i_C = 1$ mA. Design the circuit so that a current of 2 mA flows through the collector and a voltage of +5 V appears at the collector.

$$I_C = I_S e^{\frac{V_{BE}}{V_T}}$$

$$\therefore V_{BE_1} = V_T \ln \frac{I_{C1}}{I_S} \quad \text{--- (i)}$$

$$V_{BE_2} = V_T \ln \frac{I_{C2}}{I_S} \quad \text{--- (ii)}$$

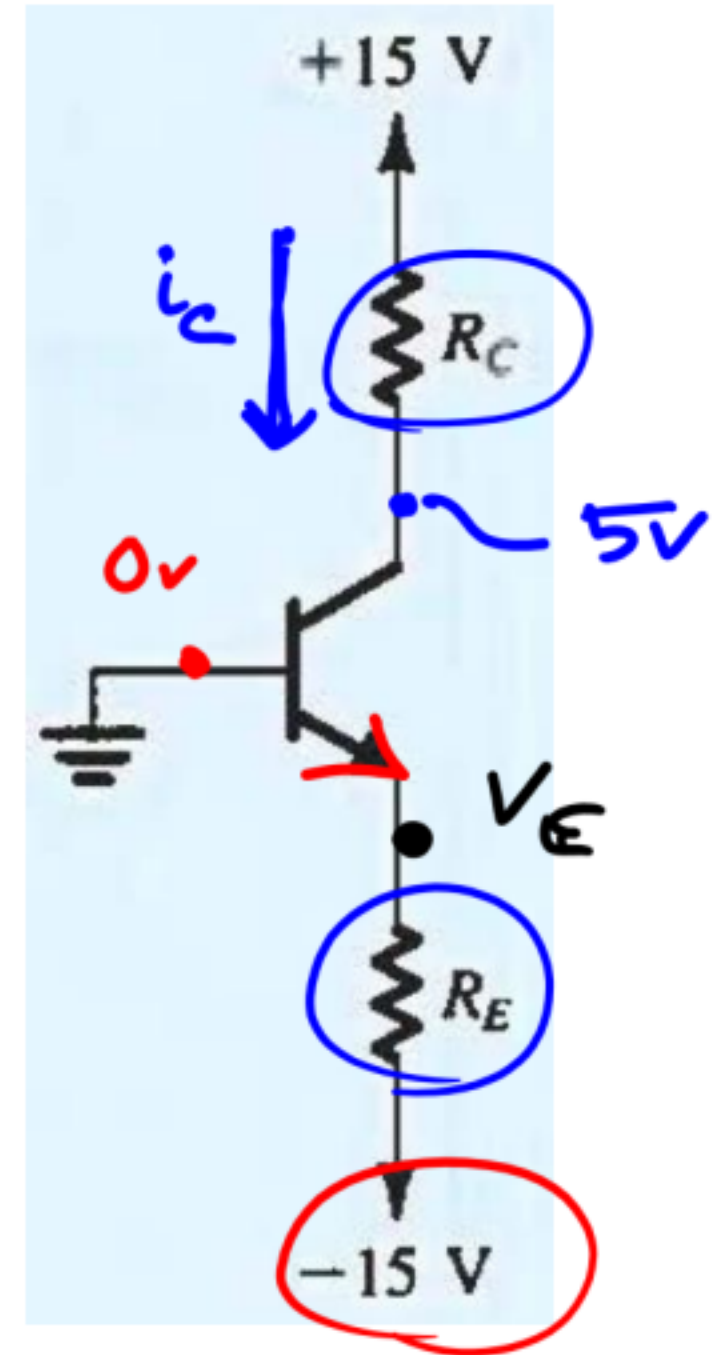
$$\begin{aligned} \text{--- (i) --- (ii)} \quad \therefore V_{BE_2} &= V_{BE_1} + V_T \ln \frac{I_{C2}}{I_{C1}} \\ &= 0.7 + 25 \text{ mV} \ln \left(\frac{2 \text{ mA}}{1 \text{ mA}} \right) \\ &= 0.72 \text{ V} \end{aligned}$$

$$\text{Now } V_{BE} = 0.72 = 0 - V_E$$

$$\therefore V_E = -0.72 \text{ V}$$

$$\begin{aligned} \therefore R_E &= \frac{V_E - (-15)}{I_E} \\ &= \frac{-0.72 + 15}{I_C \left(\frac{\beta + 1}{\beta} \right)} \\ &= 7.07 \text{ k}\Omega \end{aligned}$$

$$R_C = \frac{15 - 5}{I_C} = 5 \text{ k}\Omega$$



$$I_C = 2 \text{ mA}$$

$$\begin{aligned} I_E &= I_C + I_B \\ &= I_C \left(1 + \frac{1}{\beta} \right) \approx I_C \end{aligned}$$

In the circuit shown in Fig. E6.14, measurement indicates V_B to be $+1.0$ V and V_E to be $+1.7$ V. What are α and β for this transistor? What voltage V_C do you expect at the collector?

$$V_C = (-10) + I_C R_C \quad \text{--- ①}$$

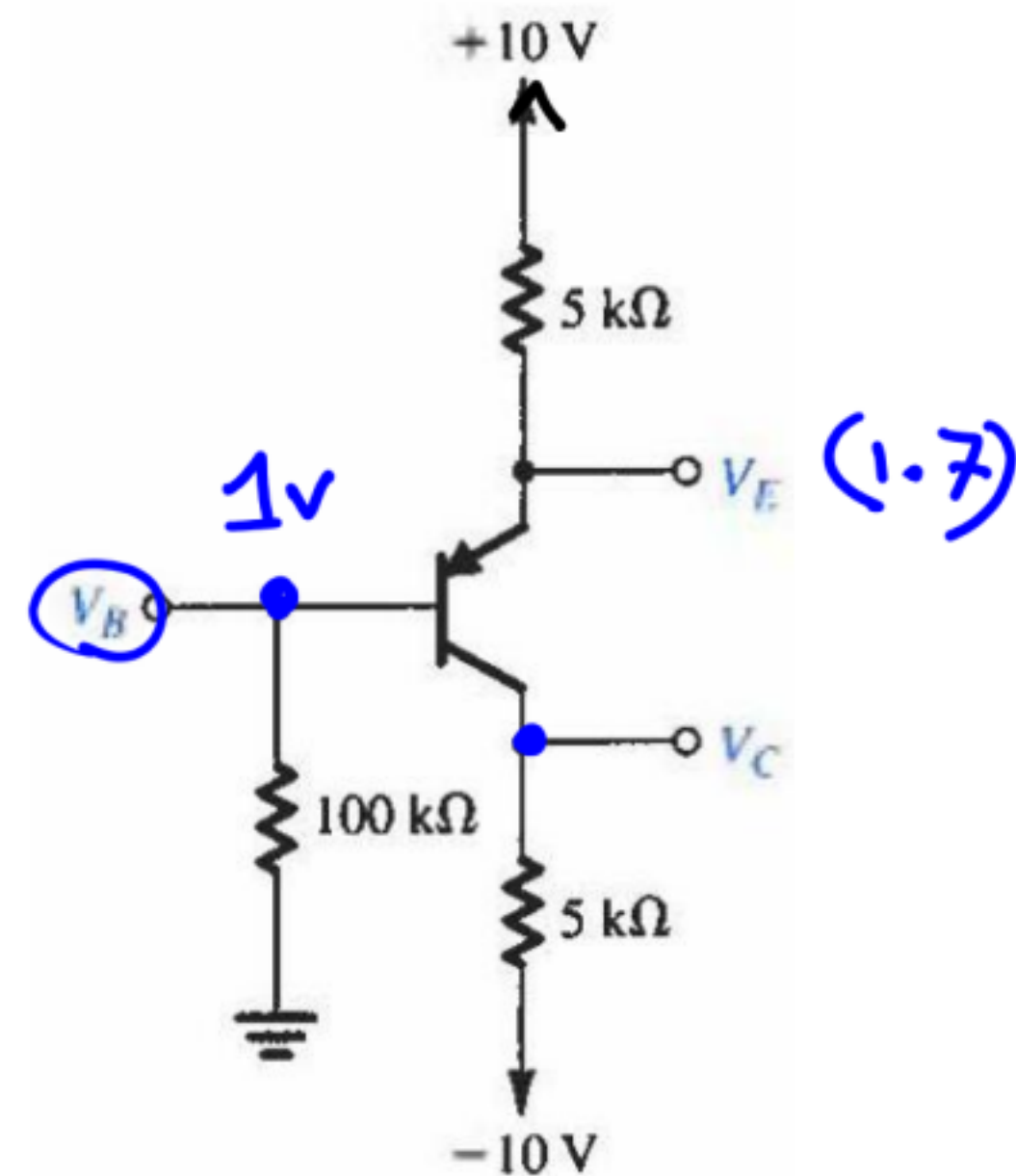
$$I_B = \frac{V_B}{100k} = 0.01 \text{ mA}$$

$$I_E = \frac{10 - V_E}{5k} = 1.66 \text{ mA}$$

$$I_C = I_E - I_B = 1.65 \text{ mA}$$

$$\beta = \frac{I_C}{I_B} = 165 \quad \alpha = \frac{I_C}{I_E} = 0.99$$

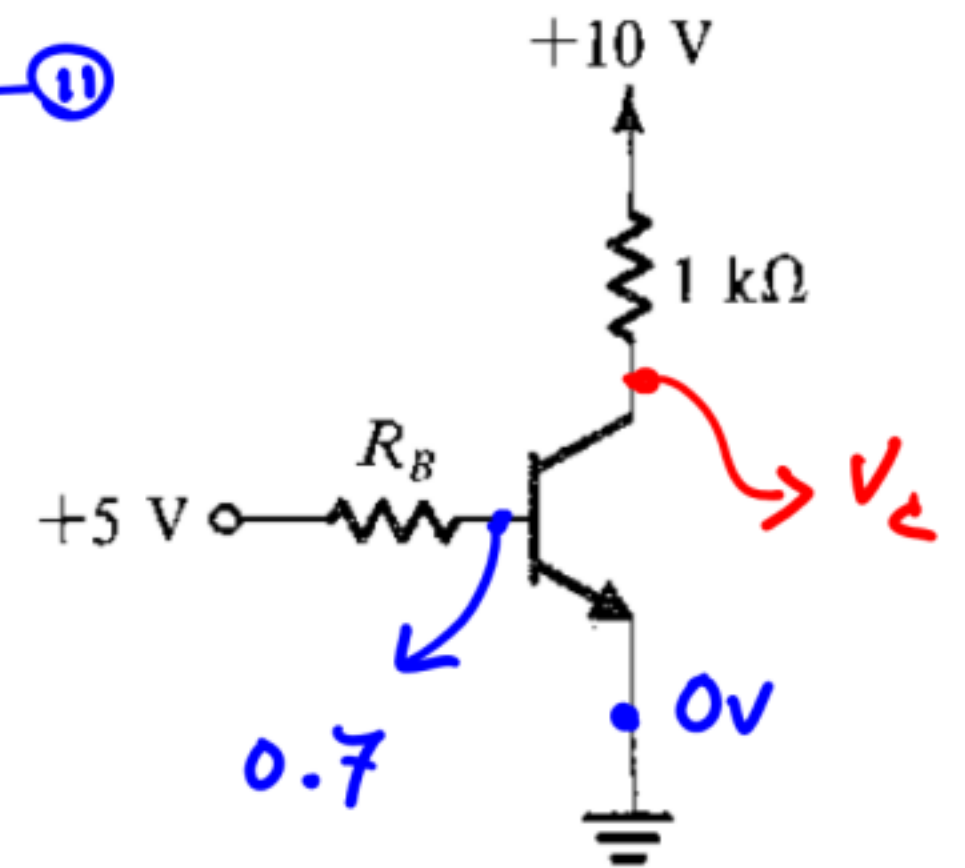
$$V_C = -10 + 1.65 \times 5 = -1.75 \text{ V}$$



The transistor in Fig. 5.33 is specified to have β in the range of (50 to 150). Find the value of R_B that results in saturation with an overdrive factor of at least 10.

$$OF = \frac{I_B(sat)}{I_B(eos)} \quad \text{--- ①} \quad \beta_{min} = 50 = \frac{I_C(sat)}{I_B(eos)} \quad \text{--- ②}$$

$$R_B = \frac{5 - 0.7}{I_B(sat)} \quad \text{--- ③}$$



from ① & ② :

$$I_B(s) = OF \times I_B(eos) = 10 \times \frac{I_C(s)}{50} = \frac{1}{5} \times 9.8 \text{ mA}$$

$$\begin{aligned} I_C(s) &= \frac{10 - V_C}{R_C} \\ &= \frac{10 - 0.2}{1\text{k}} \\ &= 9.8 \text{ mA} \end{aligned}$$

$$\therefore R_B = \frac{4.3}{9.8/5} = 2.2 \text{ k}\Omega$$



For the circuit in Fig. 6.22, determine the value of the voltage V_{BB} that results in the transistor operating

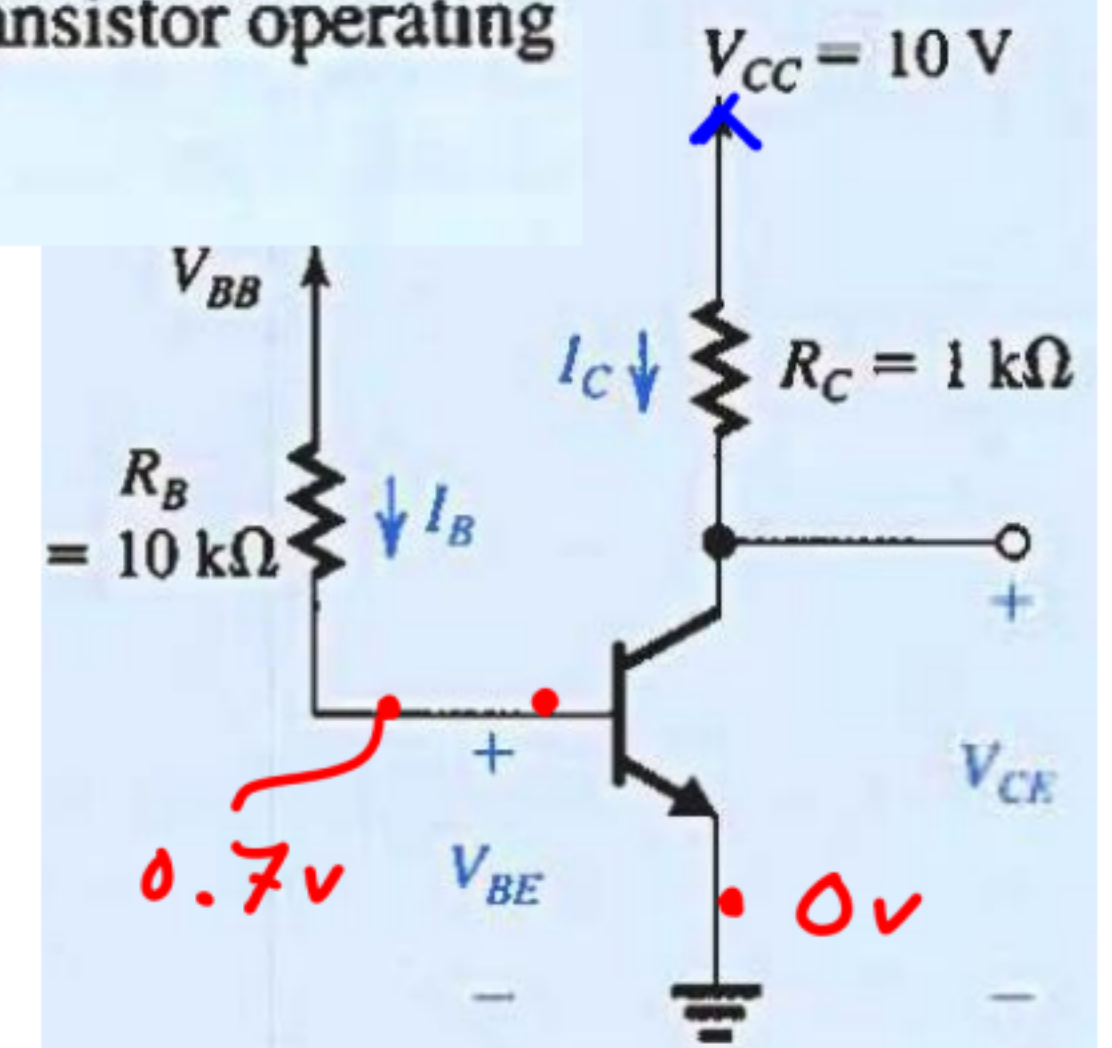
(a) in the active mode with $V_{CE} = 5\text{ V}$

$$V_{BB} = V_B + I_B R_B \text{ --- ①}$$

$$I_B = \frac{I_C}{\beta} = \left(\frac{V_{CC} - V_C}{R_C} \right) \times \frac{1}{\beta}$$

$$= \left(\frac{10 - 5}{1\text{ k}\Omega} \right) \frac{1}{50} = 0.1\text{ mA}$$

$$\therefore V_{BB} = 0.7 + 0.1 \times 10 \\ = 1.7\text{ V}$$



$$\beta = 50$$

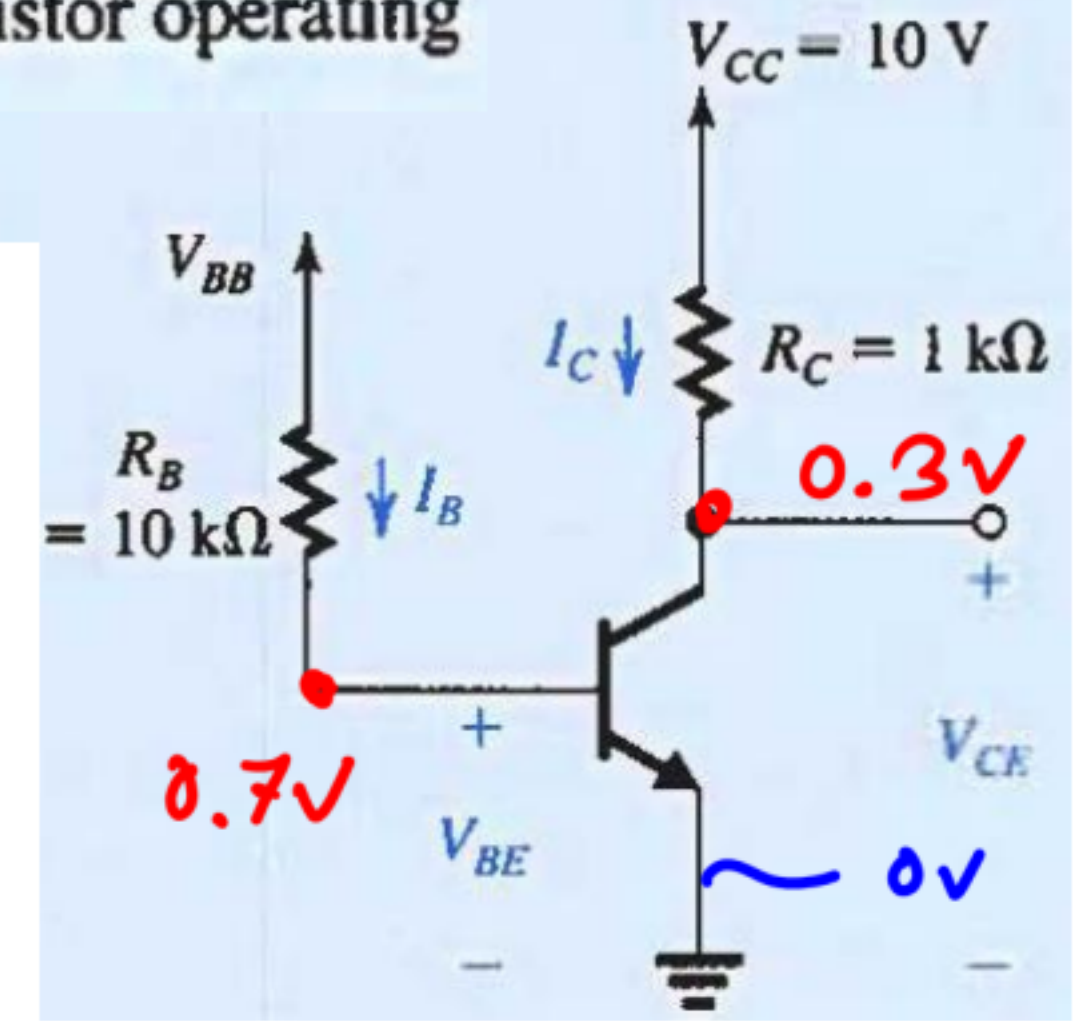
$$V_{CE} = 5\text{ V} = V_C - V_E \\ \therefore V_C = 5 + V_E = 5\text{ V}$$



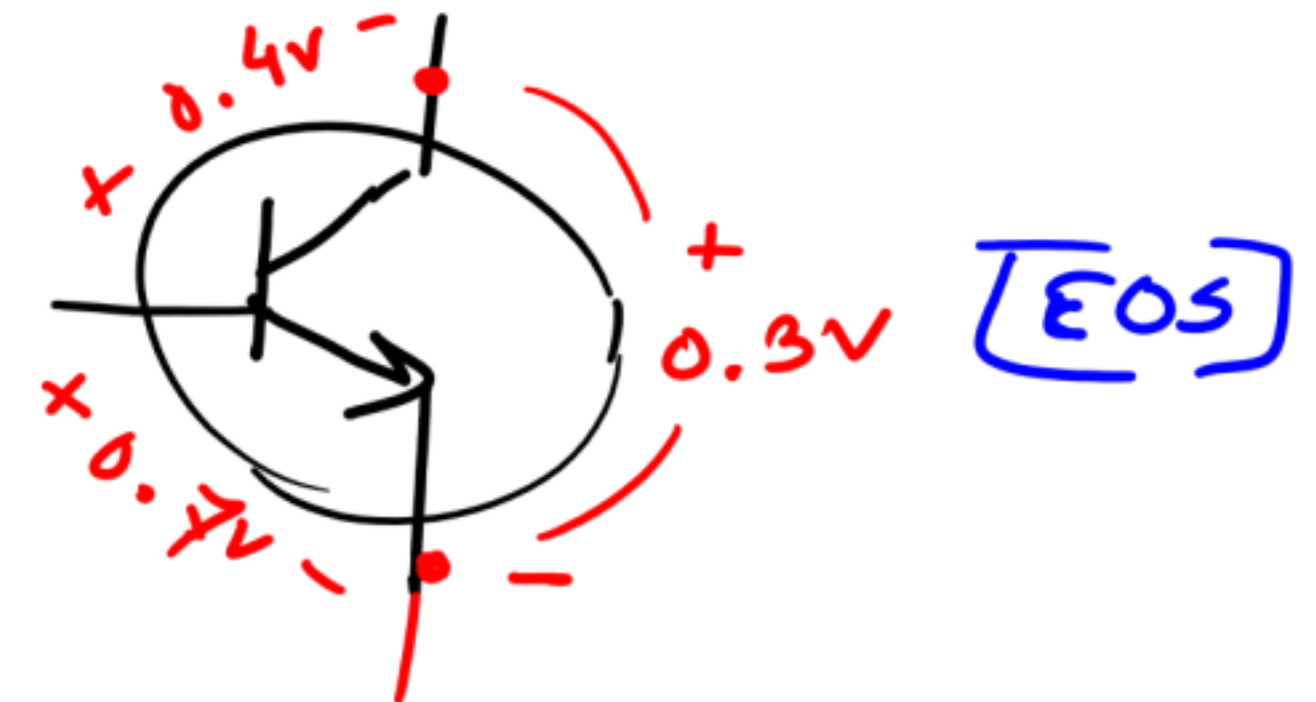
For the circuit in Fig. 6.22, determine the value of the voltage V_{BB} that results in the transistor operating

(b) at the edge of saturation

$$\begin{aligned}
 V_{BB} &= 0.7 + I_B \times 10k \\
 &= 0.7 + \frac{I_C}{\beta} \times 10k \\
 &= 0.7 + \frac{10k}{80} \left(\frac{V_{CC} - 0.3}{1k} \right) \\
 &= 1.91 \text{ V}
 \end{aligned}$$



$$\beta = 50$$



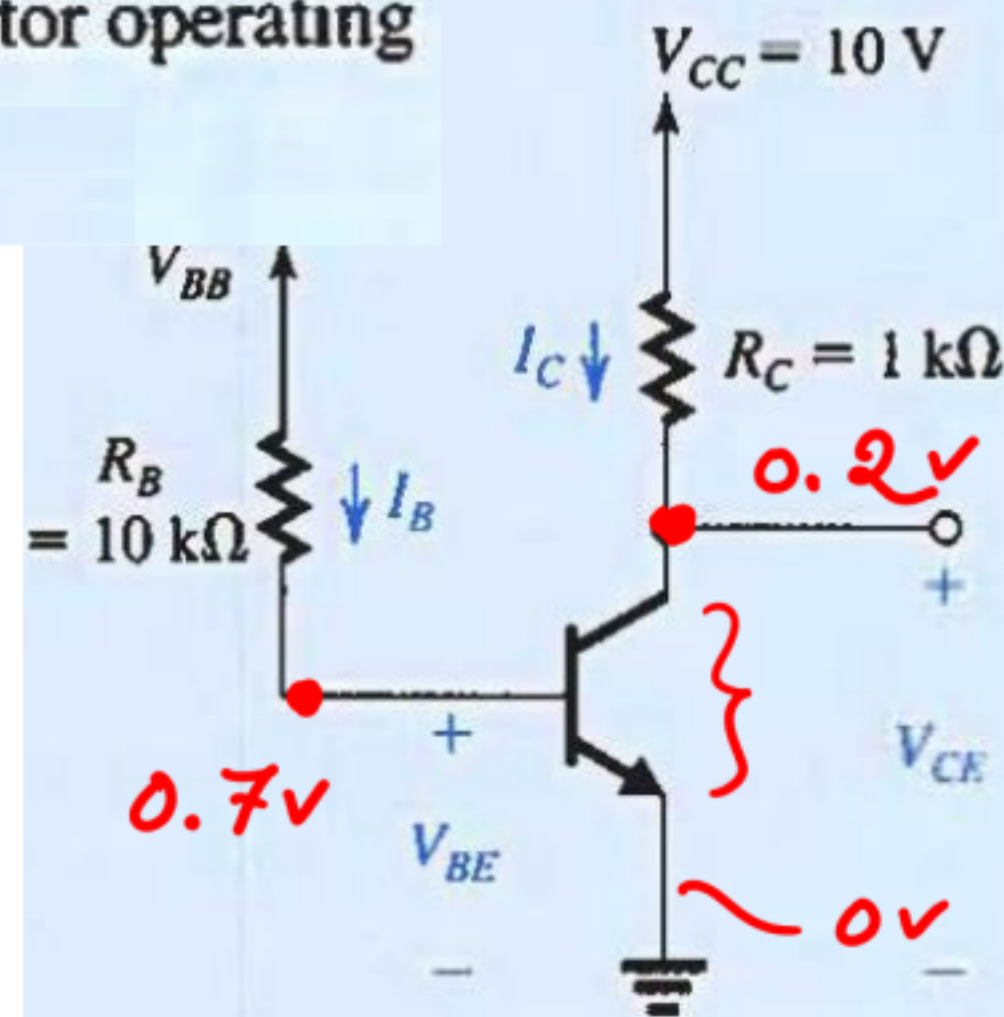
For the circuit in Fig. 6.22, determine the value of the voltage V_{BB} that results in the transistor operating

(c) deep in saturation with $\beta_{\text{forced}} = 10$

$$\beta_{\text{forced}} = 10 = \frac{I_C(s)}{I_B(s)}$$

$$V_{BB} = 0.7 + I_B(s) R_B$$
$$= 0.7 + 10\text{K} \times \frac{I_C(s)}{\beta_f}$$

$$= 0.7 + \frac{10\text{K}}{10} \times \left(\frac{V_{CC} - 0.2}{1\text{K}} \right) = 10.5\text{V}$$



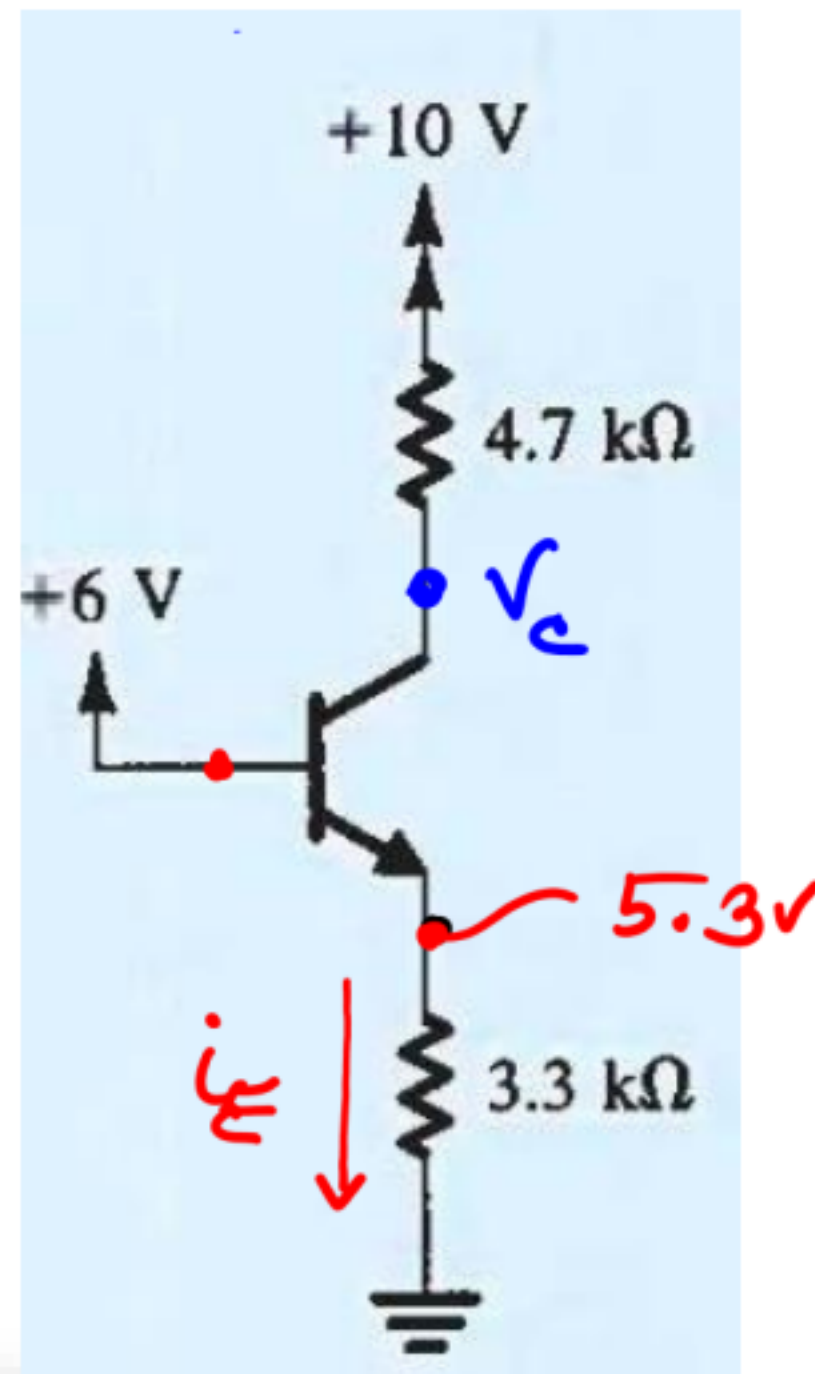
$$\beta = 50$$



We want to analyze the circuit of Fig. 6.24(a) to determine the voltages at all nodes and the currents through all branches. Note that this circuit is identical to that of Fig. 6.23 except that the voltage at the base is now +6 V. Assume that the transistor β is specified to be *at least* 50.

$\rightarrow \beta, \alpha, \epsilon$

Let: active



$$\bullet \hat{I}_E = \frac{5.3}{3.3k} = 1.61 \text{ mA} \quad \bullet \hat{I}_c = \alpha \hat{I}_E = \frac{50}{51} \times 1.61 = 1.57 \text{ mA}$$

$$\bullet \hat{I}_B = \hat{I}_E - \hat{I}_c = 0.04 \text{ mA}$$

$$V_B = 6 \text{ V}$$

$$V_c = 10 - \hat{I}_c R_c$$

$$= 10 - 1.57 \times 4.7$$

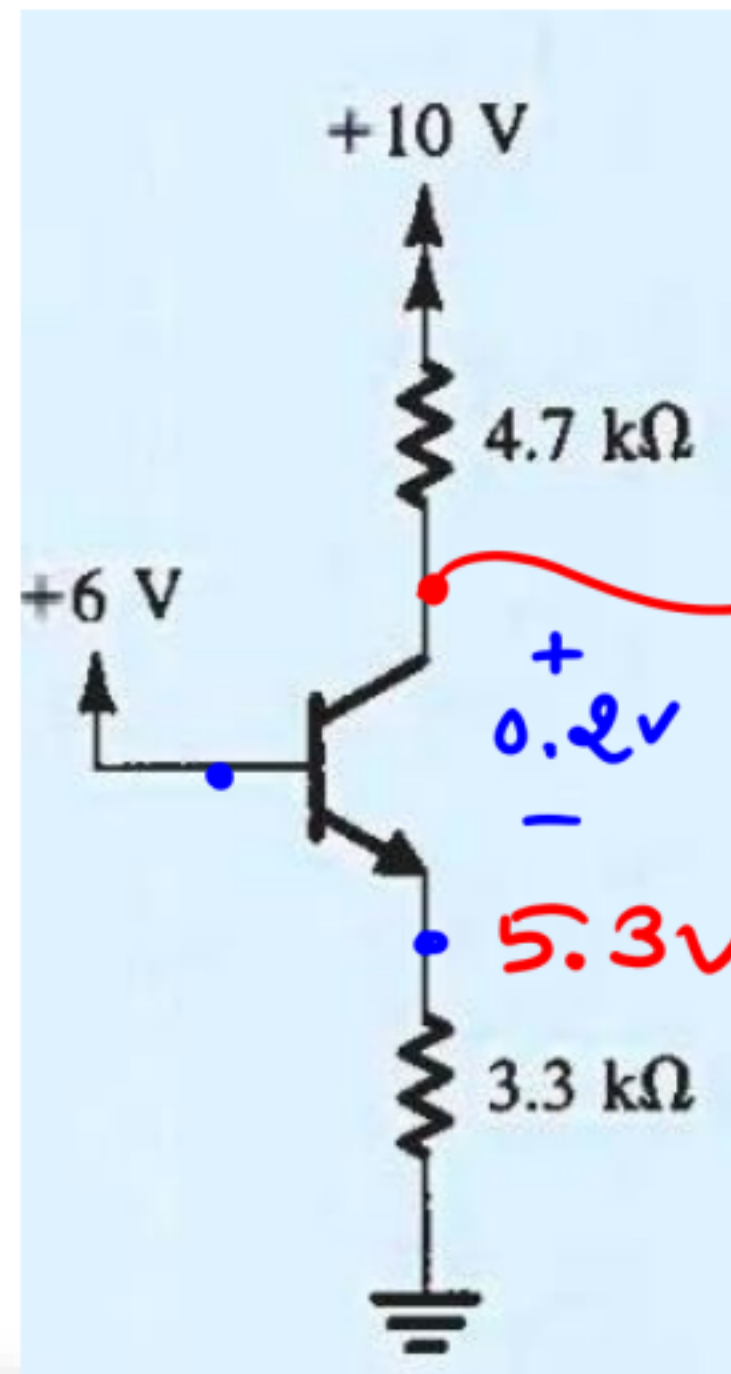
$$= 2.62 \text{ V}$$

$$(\beta = 50) \rightarrow \alpha = \frac{\beta}{1 + \beta}$$

$$V_{BC} = 6 - 2.62 = 3.38 > 0.4 \text{ V}$$

Not active !

We want to analyze the circuit of Fig. 6.24(a) to determine the voltages at all nodes and the currents through all branches. Note that this circuit is identical to that of Fig. 6.23 except that the voltage at the base is now +6 V. Assume that the transistor β is specified to be *at least* 50.



saturation

$$V_C = 5.3 + 0.2 = 5.5V$$

$$V_E = 5.3V$$

$$V_B = 6V$$

$$I_C = \frac{10 - 5.5}{4.7k}$$

$$I_E = \frac{5.3V}{3.3k}$$

$$I_B = I_E - I_C$$



We want to analyze the circuit of Fig. 6.26(a) to determine the voltages at all nodes and the currents through all branches.

active mode (let)

$$\bullet I_E = \frac{V^+ - 0.7}{2K} = 4.65 \text{ mA}$$

$$\bullet I_B = \frac{I_E}{\beta + 1} = 0.046 \text{ mA}$$

$$\bullet I_C = I_E - I_B = 4.6 \text{ mA}$$

$$\begin{aligned} I_E &= I_C + I_B \\ &= \beta I_B + I_B \\ &= I_B (\beta + 1) \\ &= I_C \times \frac{\beta}{1 + \beta} \end{aligned}$$

$$V_C = V^- + I_C R_C$$

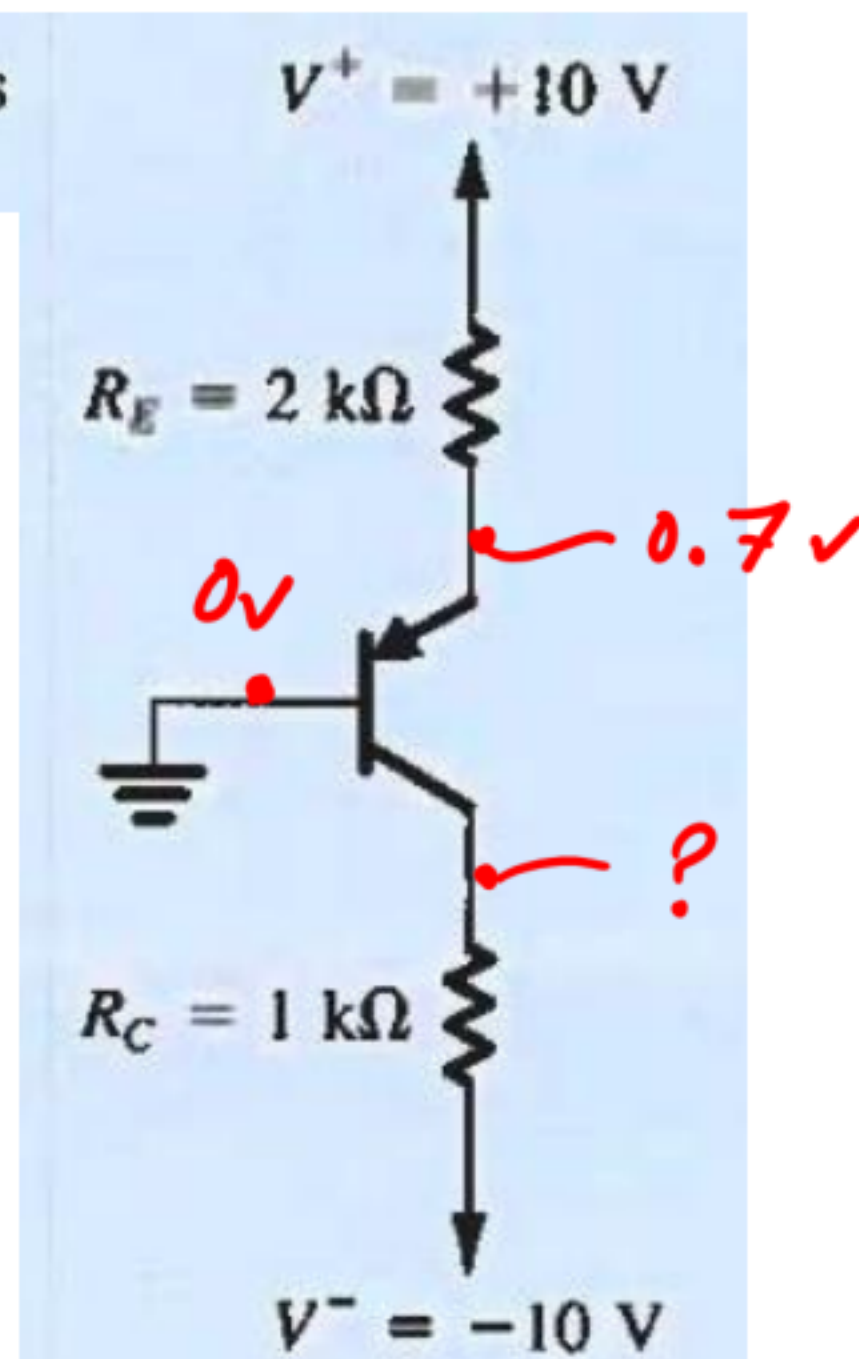
$$= (-10) + 4.6 \times 1 = -5.4 \text{ V}$$

$$V_{CB} = -5.4 < 0.4$$

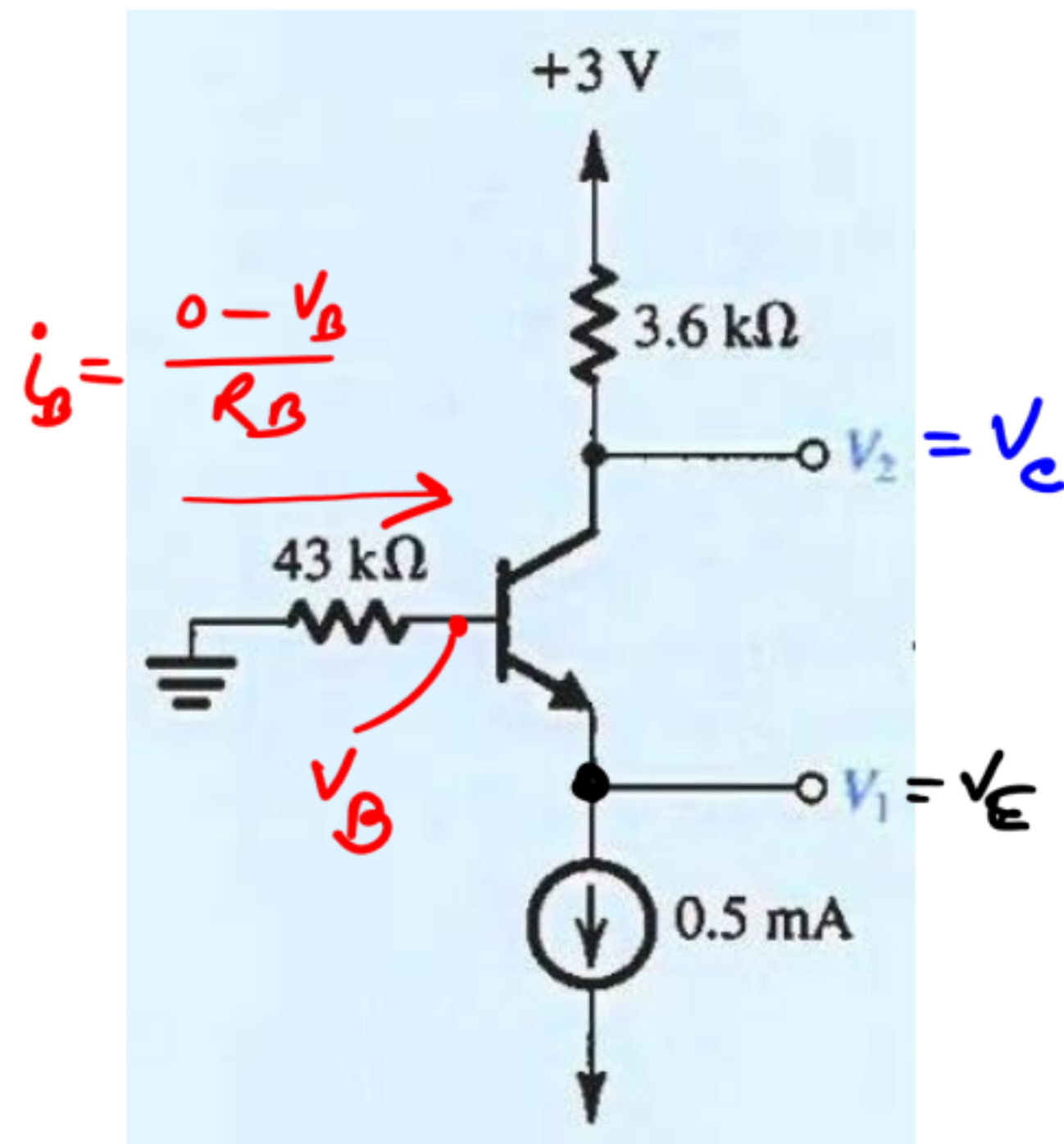
$\therefore$ active

$$\bullet \beta = 100$$

(let)



Find V_1 & V_2



$$I_E = 0.5 \text{ mA}$$

$$i_c = \frac{\beta}{1+\beta} \times 0.5$$

$$= \frac{0.5 \times 100}{101}$$

$$= 0.495 \text{ mA}$$

$$i_B = 0.005 \text{ mA}$$

$$V_2 = 3 - I_c R_c = 1.22 \text{ V}$$

$$= V_c$$

$$V_{BE} = 0.7 \text{ V}$$

$$\Rightarrow V_B - V_1 = 0.7$$

$$\therefore V_1 = V_B - 0.7$$

$$= (0 - I_B R_B) - 0.7$$

$$= -0.91 \text{ V}$$

$$V_B = -I_B R_B$$

$$= -0.22 \text{ V}$$

$(\beta = 100)$
 → By default; if not given

$$V_{BC} = -1.44 < 0.4$$

For the circuit in Fig. 6.26(a), find the largest value to which R_C can be raised while the transistor remains in the active mode.

$$R_C = \frac{V_C - V^-}{I_C}$$

$$\therefore R_C(\max) = \frac{V_C(\max) - (-10)}{4.604 \text{ mA}}$$

$$R_C(\max) = \frac{0.4 + 10}{4.604 \text{ mA}} \\ = 2.26 \text{ k}\Omega$$

$$I_C = \alpha I_E$$

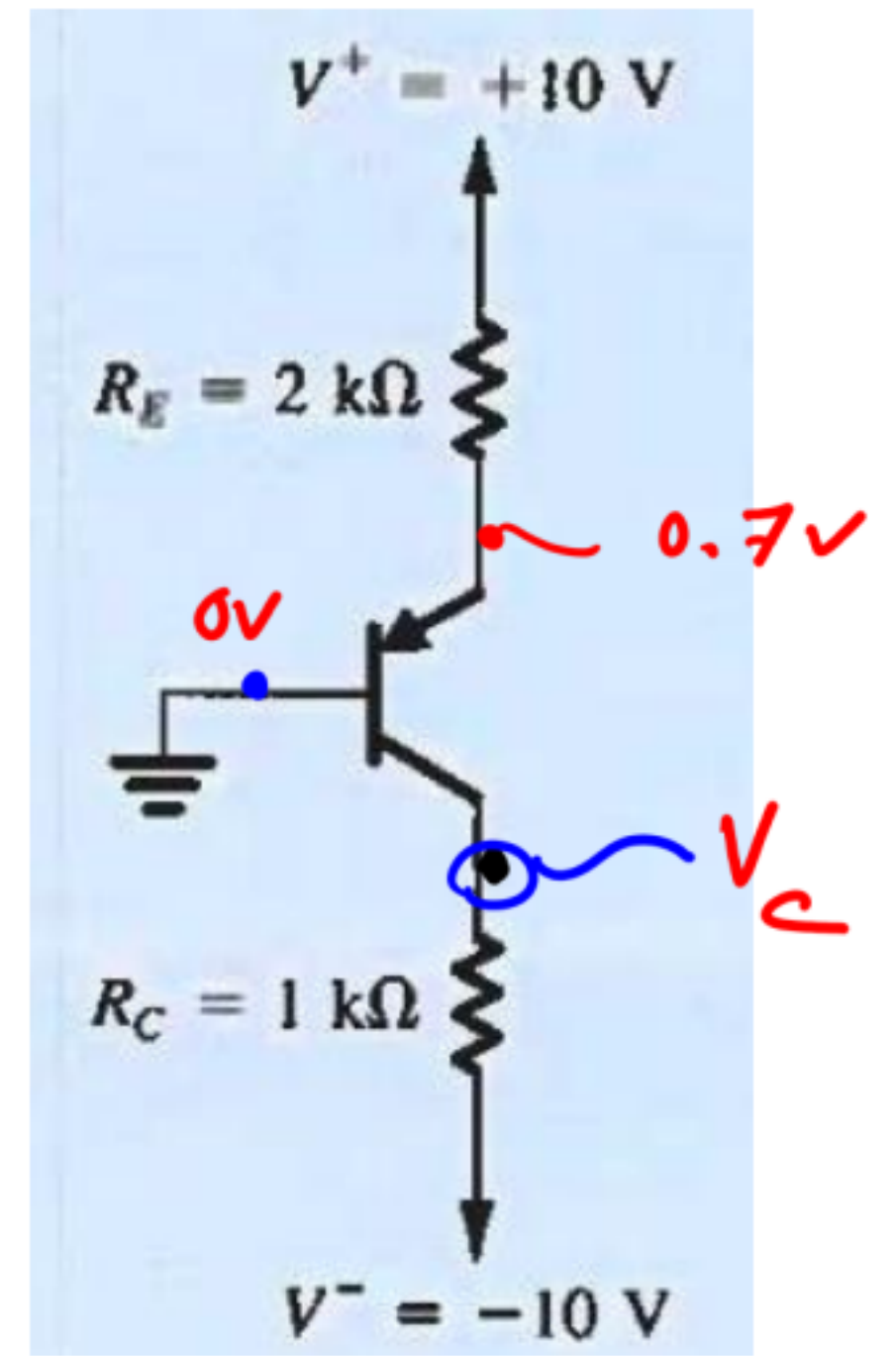
$$= \frac{100}{101} \times \frac{10 - 0.7}{2 \text{ k}}$$

$$= 4.604 \text{ mA}$$

$$V_{CB} < 0.4 \text{ (active)}$$

$$\therefore V_C < V_B + 0.4 = 0.4$$

$$\boxed{V_C < 0.4} \therefore V_C(\max) = 0.4 \text{ V}$$



$$(\beta = 100)$$

$$\therefore \alpha = \frac{100}{101}$$



We want to analyze the circuit of Fig. 6.28(a) to determine the voltages at all nodes and the currents through all branches. The minimum value of β is specified to be 30.

$$V_E = V_B + 0.7 \quad \text{--- ①}$$

$$\# \dot{I}_E = (\beta + 1) \dot{I}_B$$

$$\Rightarrow \frac{5 - V_E}{1k} = 31 \times \frac{V_B}{10k}$$

$$\Rightarrow \frac{5 - V_B - 0.7}{1} = 3.1 V_B \quad \therefore V_C = -5 + I_C R_C = 20.75V !$$

$$\Rightarrow V_B = 1.7V$$

$$\therefore V_E = 2.4V$$

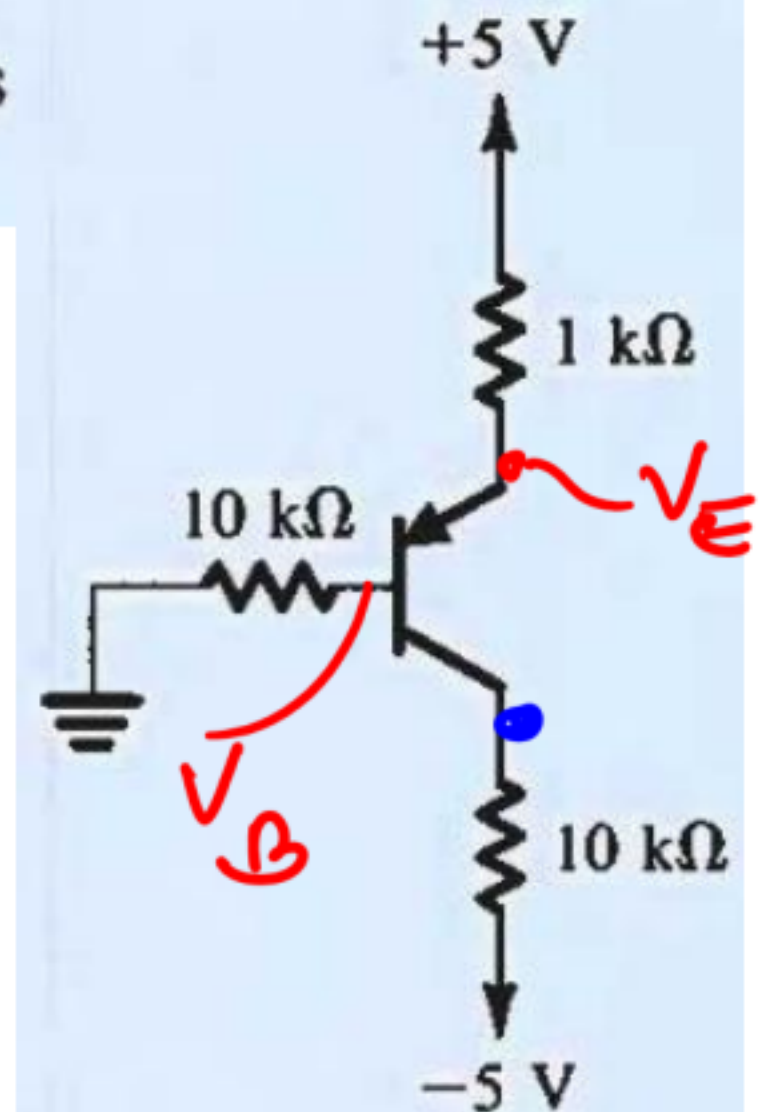
$$\dot{I}_E = \frac{5 - 2.4}{1k} = 2.6mA$$

$$\dot{I}_B = \frac{\dot{I}_E}{101} = 0.025mA$$

$$\dot{I}_C = 2.575mA$$

$$\# V_{CB} = 19.05V > 0.4V$$

Not in active mode



We want to analyze the circuit of Fig. 6.28(a) to determine the voltages at all nodes and the currents through all branches. The minimum value of β is specified to be 30.

<saturation mode>

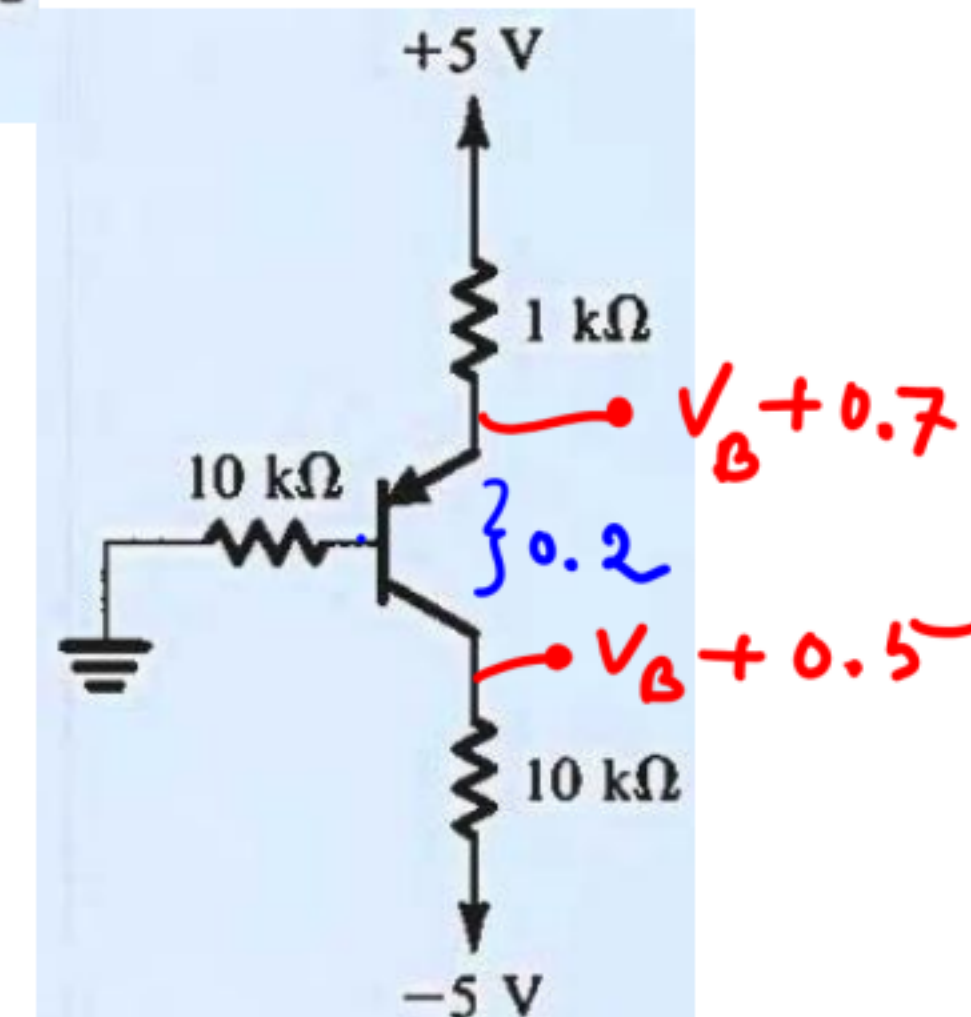
$$i_E = i_B + i_C$$

$$\Rightarrow \frac{5 - V_E}{1K} = \frac{V_B - 0}{10K} + \frac{V_C - (-5)}{10K}$$

$$\Rightarrow \underbrace{\frac{5 - V_B - 0.7}{1}}_{i_E} = \underbrace{\frac{V_B}{10}}_{i_B} + \underbrace{\frac{V_B + 0.5 + 5}{10}}_{i_C} \rightarrow V_B = 3.125V$$

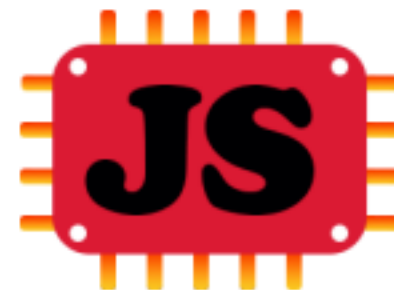
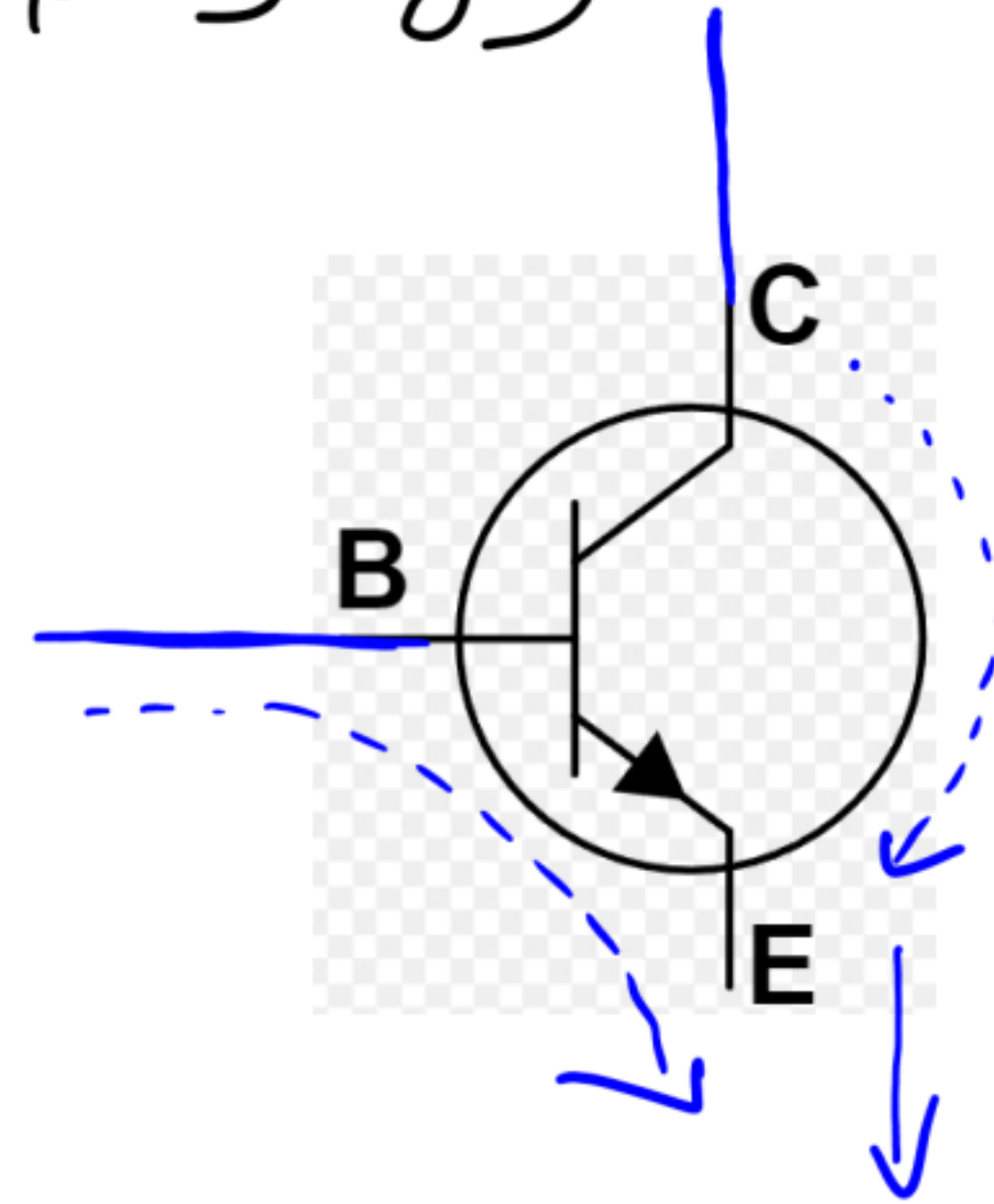
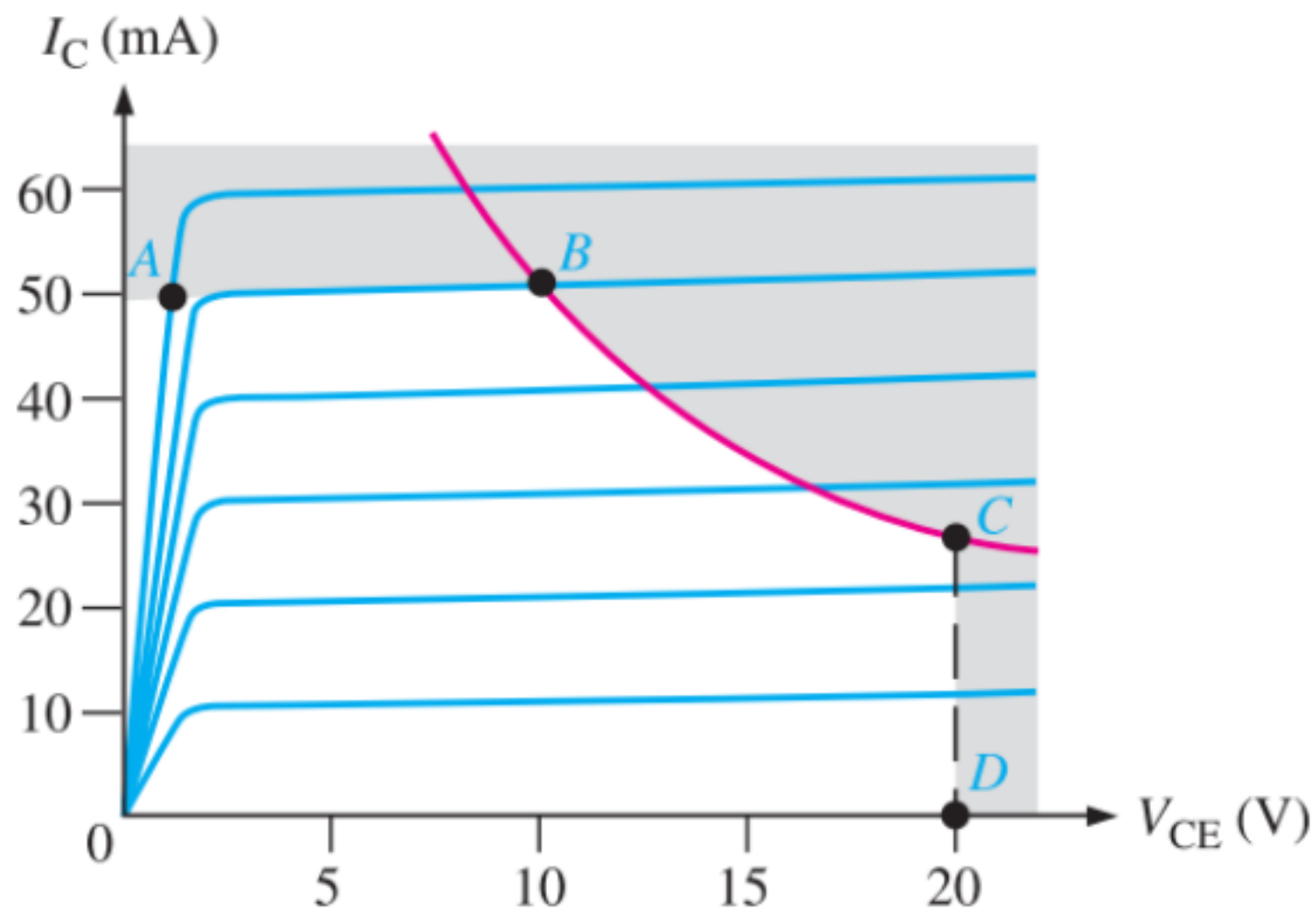
$$\therefore V_E = 3.825V$$

$$\therefore V_C = 3.625V$$

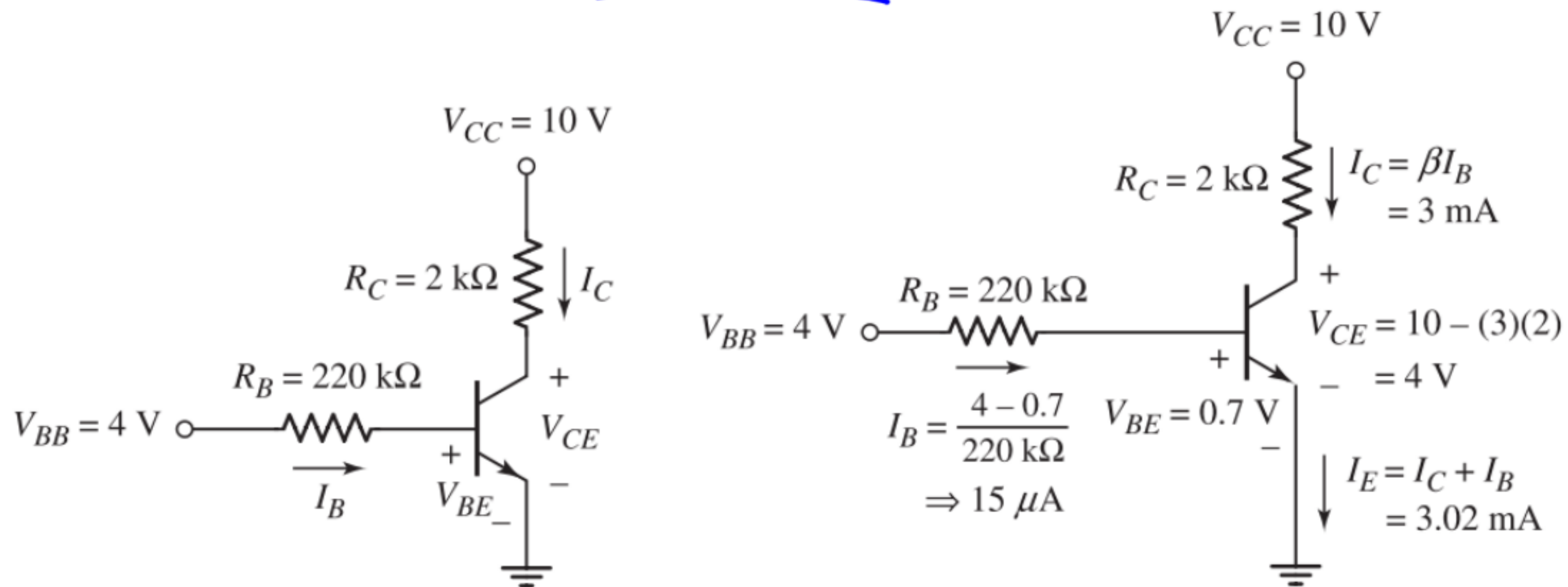


Maximum Power Dissipation in a Transistor

$$P = V_{BE}I_B + V_{CE}I_C \approx I_C V_{CE} \quad (\text{if } \beta \text{ very high})$$



Objective: Calculate the base, collector, and emitter currents and the C-E voltage for a common-emitter circuit. Calculate the transistor power dissipation.



The power dissipated in the transistor is found to be

$$P_T = I_B V_{BE}(\text{on}) + I_C V_{CE} = (0.015)(0.7) + (3)(4) \cong I_C V_{CE}$$



A certain transistor is to be operated with $V_{CE} = 6$ V. If its maximum power rating is 250 mW, what is the most collector current that it can handle?

$$P = I_B V_{BE} + I_C V_{CE} \approx I_C V_{CE}$$

$$\therefore P(\max) = I_C(\max) \cdot V_{CE}$$

$$\therefore I_C(\max) = \frac{250 \text{ mW}}{6 \text{ V}} = 41.67 \text{ mA}$$



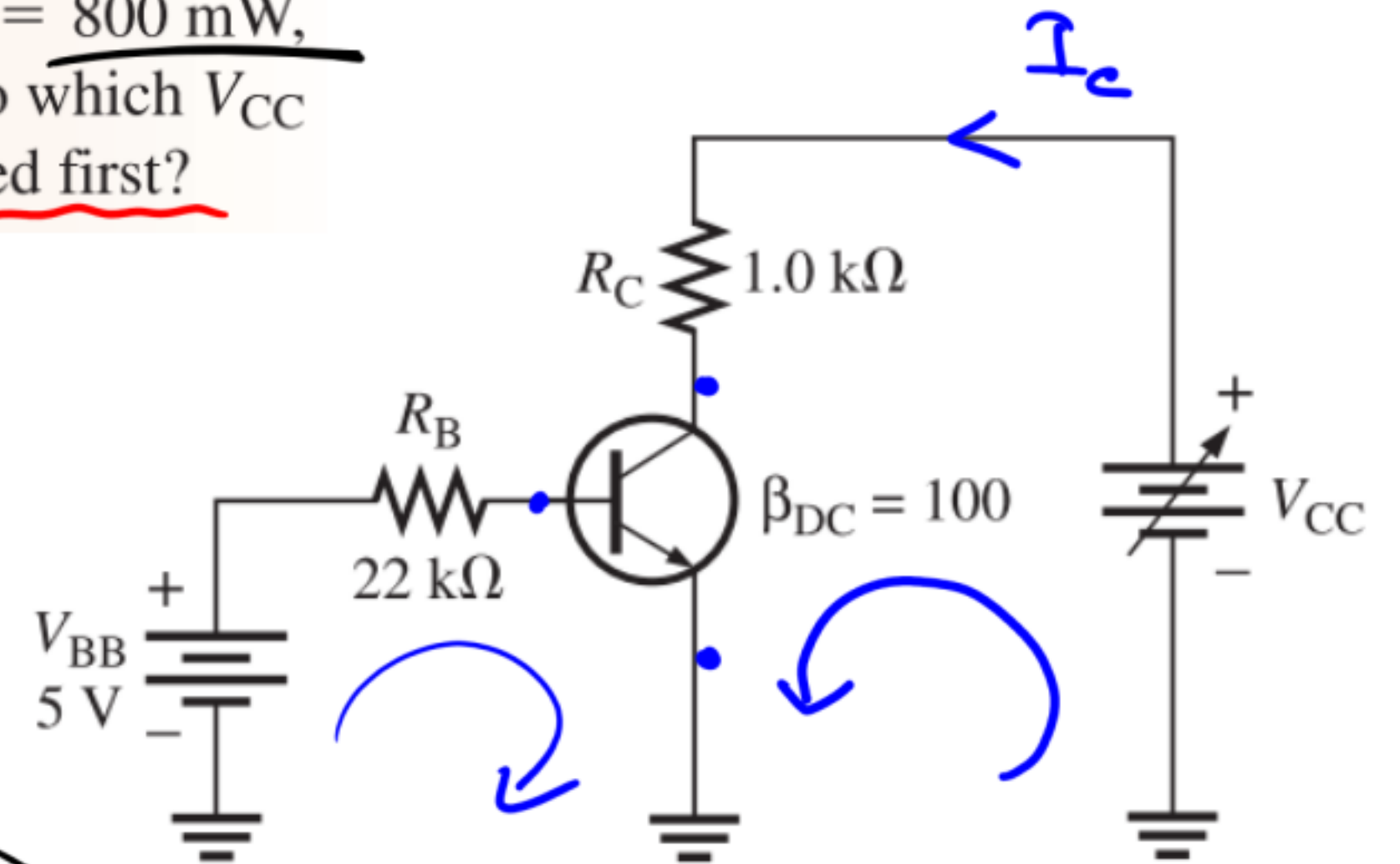
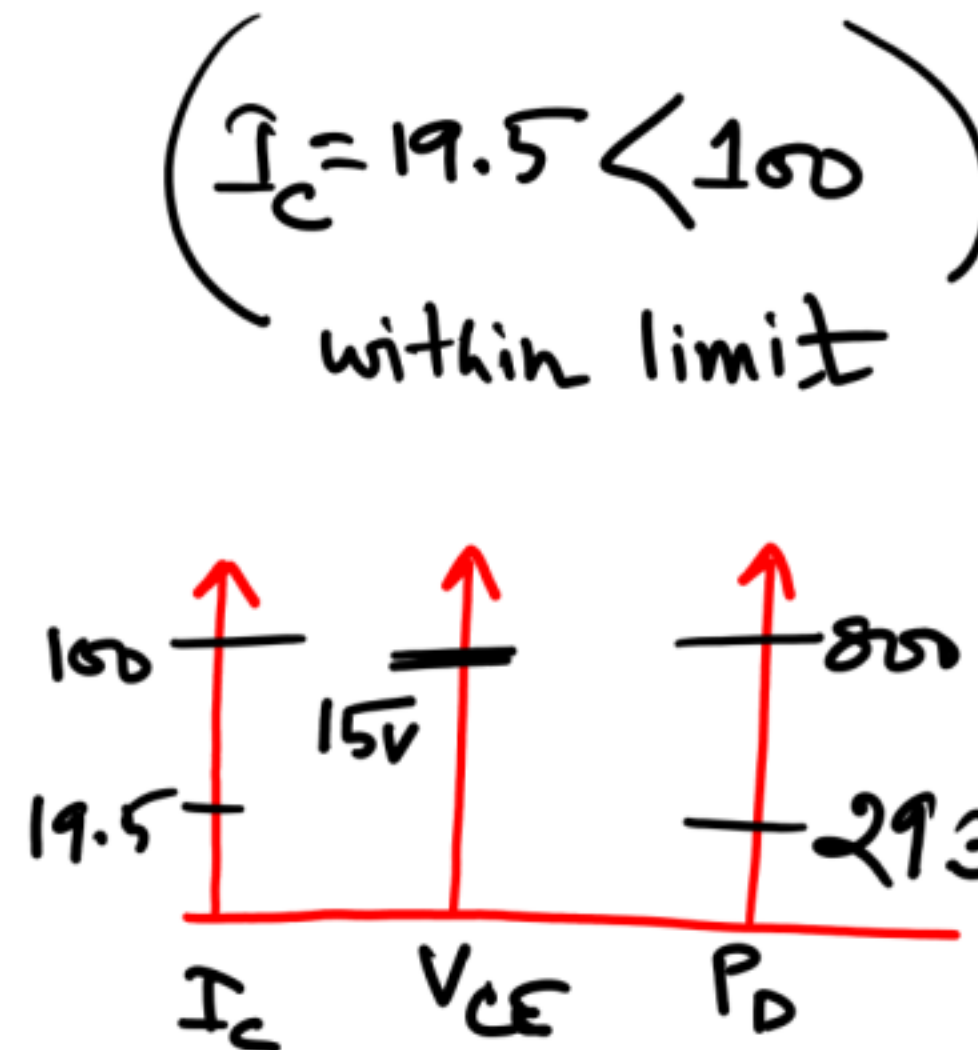
The transistor in Figure 4-19 has the following maximum ratings: $P_{D(max)} = 800 \text{ mW}$, $V_{CE(max)} = 15 \text{ V}$, and $I_{C(max)} = 100 \text{ mA}$. Determine the maximum value to which V_{CC} can be adjusted without exceeding a rating. Which rating would be exceeded first?

$$V_{CC} = V_{CE} + I_C R_C \quad \text{--- ①}$$

$$\begin{aligned} \therefore V_{CE(max)} &= V_{CE(max)} + I_C R_C \\ &= 15 + 19.5 \times 1 \\ &= 34.5 \text{ V} \quad \text{[Ans.]} \end{aligned}$$

$$\begin{aligned} P_D(max) &= I_C V_{CE(max)} \\ &= 19.5 \text{ mA} \times 15 \text{ V} \\ &= 293 \text{ mW} \end{aligned}$$

$\therefore V_{CE}$ will be exceeded first.



$$\begin{aligned} I_B &= \frac{V_{BB} - V_B}{R_B} \\ &= \frac{5 - 0.7}{22 \text{ k}} \end{aligned}$$

$$\therefore I_C = \beta I_B = 19.5 \text{ mA}$$

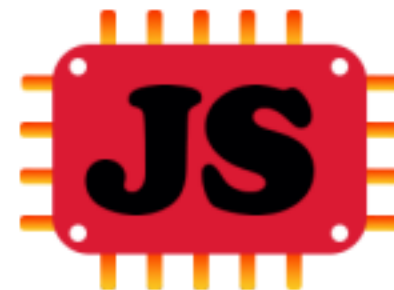


Derating P_D (max)

- $P_D(max)$ is usually specified at 25°C . For higher temperatures, it is less.
- Derating factors are mentioned in transistor data-sheets
- Derating factor, $D = 5 \frac{\text{mW}}{^\circ\text{C}}$ (for example)

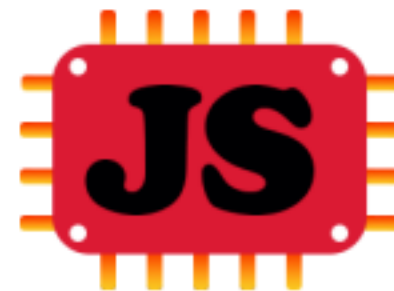
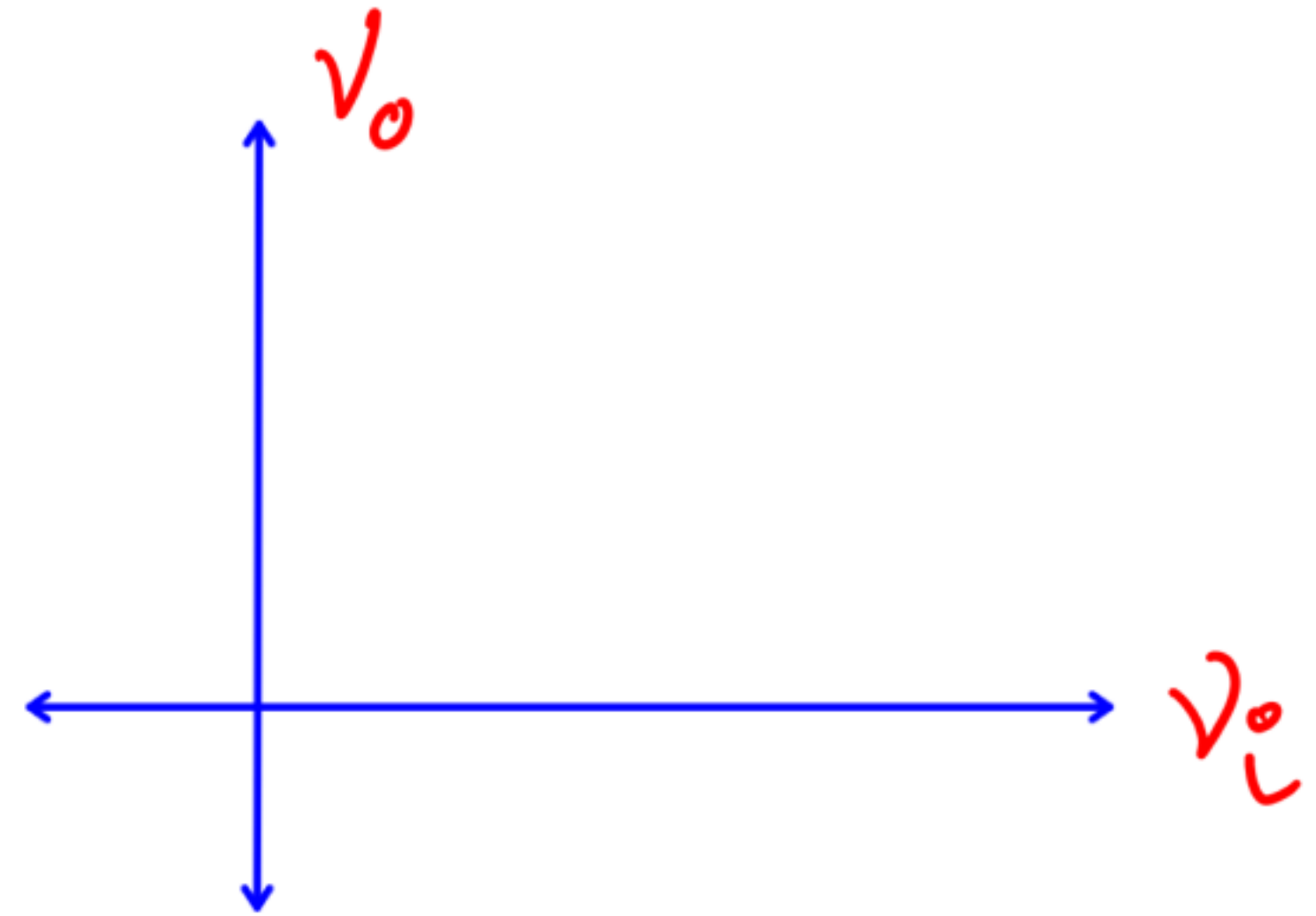
A certain transistor has a $P_{D(max)}$ of 1 W at 25°C . The derating factor is $5 \text{ mW}/^\circ\text{C}$. What is the $P_{D(max)}$ at a temperature of 70°C ?

$$\begin{aligned} P_H &= P_L - D \times \Delta\theta \\ &= 1\text{W} - \left(\frac{5 \times 10^{-3}}{1^\circ\text{C}} \times (70 - 25)^\circ\text{C} \right) \\ &= 0.775\text{W} \end{aligned}$$



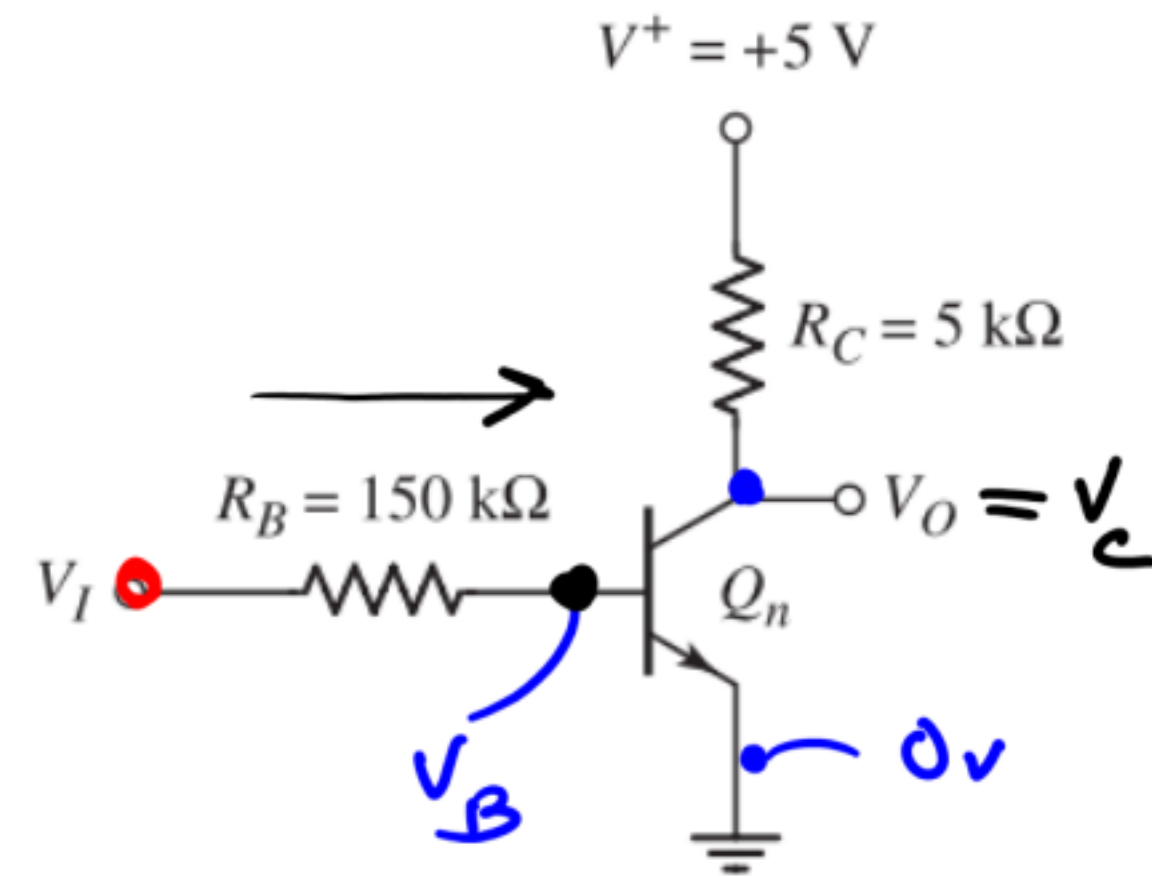
Voltage Transfer Characteristics

- Voltage-transfer characteristic curve is found by plotting output voltage against input voltage.
- It is fundamental in designing and understanding amplifier circuits.



Find the voltage transfer curve for the following circuit:

Assume npn transistor parameters of $V_{BE}(\text{on}) = 0.7 \text{ V}$, $\beta = 120$, $V_{CE}(\text{sat}) = 0.2 \text{ V}$, and $V_A = \infty$, and pnp transistor parameters of $V_{EB}(\text{on}) = 0.7 \text{ V}$, $\beta = 80$, $V_{EC}(\text{sat}) = 0.2 \text{ V}$, and $V_A = \infty$.



$$V_{BE} < 0.7 \rightarrow \text{cut-off}$$

$$\therefore V_i < 0.7 \rightarrow \text{cut-off} \therefore V_o = 5\text{V}$$

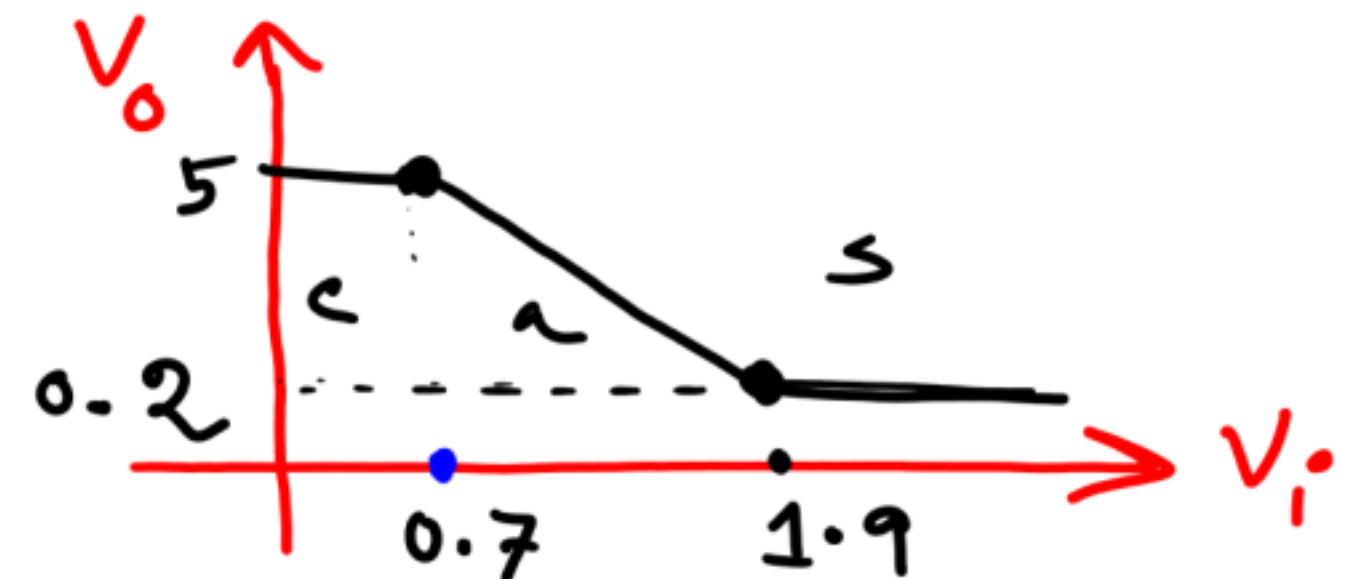
$$V_i > 0.7 \rightarrow \text{active} \rightarrow \text{saturation}$$

$$V_o = 5 - i_c R_c = 5 - \beta i_b R_c = 5 - \beta R_c \left(\frac{V_i - 0.7}{150\text{k}} \right) = 5 - 4(V_i - 0.7)$$

$$V_o = 7.8 - 4V_i$$

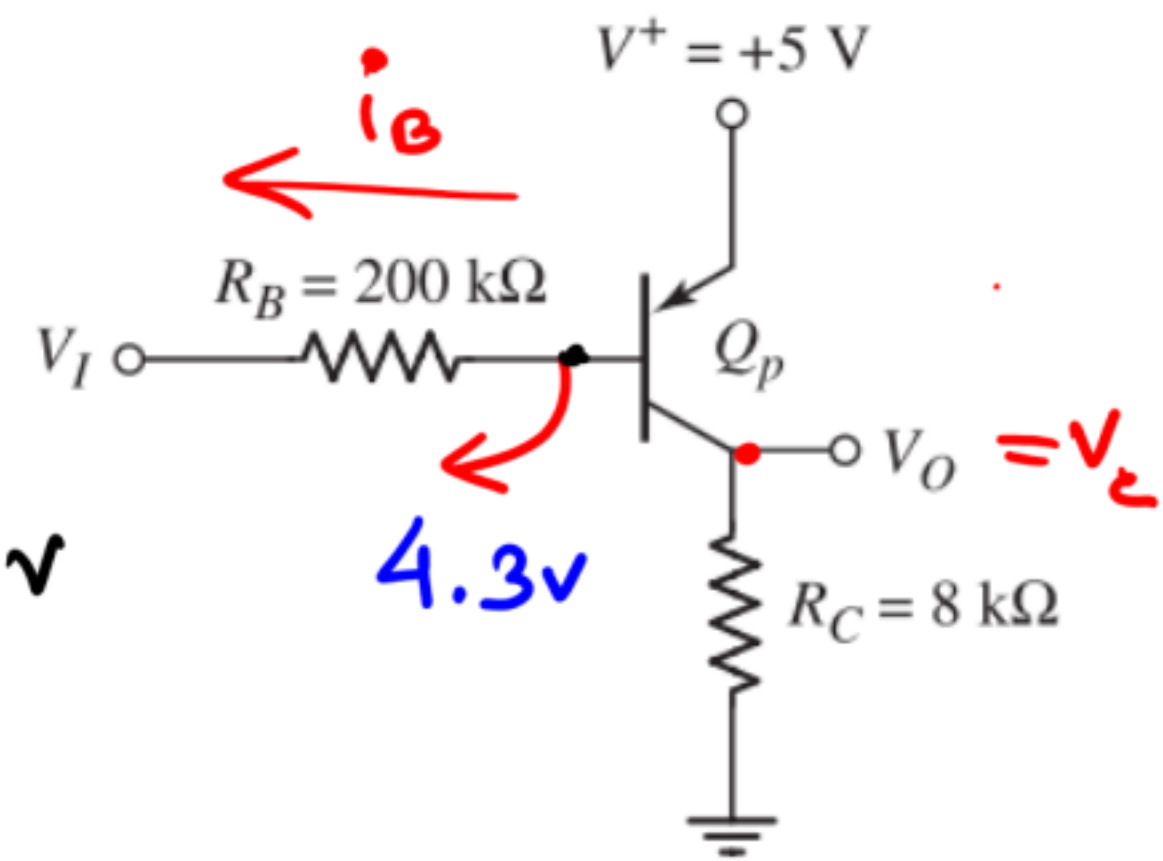
saturation: $V_{CE} = 0.2\text{V}$
 $\therefore V_o = 0.2\text{V}$

$$V_i = 1.9\text{V}$$



Find the voltage transfer curve for the following circuit:

Assume npn transistor parameters of $V_{BE}(\text{on}) = 0.7 \text{ V}$, $\beta = 120$, $V_{CE}(\text{sat}) = 0.2 \text{ V}$, and $V_A = \infty$, and pnp transistor parameters of $V_{EB}(\text{on}) = 0.7 \text{ V}$, $\beta = 80$, $V_{EC}(\text{sat}) = 0.2 \text{ V}$, and $V_A = \infty$.



$$V_{EB} < 0.7 \rightarrow \text{cut-off} \rightarrow 5 - V_B < 0.7 \Rightarrow V_B > 4.3 \text{ V}$$

$$\therefore V_i > 4.3 \text{ V} \rightarrow \text{cut-off} \rightarrow V_o = 0 \text{ V}$$

$$V_i < 4.3 \text{ V} \rightarrow \text{active} \rightarrow \text{saturation}$$

$$V_o = i_c R_c = \beta i_b R_c = \beta \left(\frac{4.3 - V_i}{200 \text{ k}} \right) \times 8 \text{ k} = 3.2 (4.3 - V_i) \quad \text{--- ①}$$

saturation ?

$$V_o = 5 - 0.2 = 4.8 \text{ V}$$

$$V_i = 2.8 \text{ V}$$

